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Electricity Supply Solutions for an Educational Center in Tanzania

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Abstract

The aim of this study was to investigate electricity supply solutions for an educational center that is being built in Chonyonyo Tanzania. Off-grid power generation solutions and further optimization possibilities were studied for the case.

The study was done for Engineers Without Borders in Sweden. Who are working with Mavuno Project on the educational center. The school is set to start operating in year 2015 with 40 girl students in the beginning. The educational center will help to improve gender equality by offering high quality education in a safe environment for girls in rural area.

It is important for the system to be economically and environmentally sustainable. The area has great potential for photovoltaic power generation. Thus PV was considered as the primary power generation and a diesel generator as a reliable backup. The system size optimization was done with HOMER. For the simulations HOMER required component data, weather data and load data. Common components were chose with standard properties, the loads were based on load estimations from year 2011 and the weather data was acquired from NASA database. The system size optimization result for this base case was a system with 26 kW PV; 5.5 kW diesel generator, 15 kW converter and 112 T-105 batteries. The initial cost of the system was 55 875 €, the total net present cost 92 121 € and the levelized cost of electricity 0.264 €/kWh.

In addition three optimization possibilities were studied. First it was studied how the system should be designed and how it would affect the system size to have night loads (security lights) use DC and could the system then be extended in blocks. As a result it was found out that the system size could be decreased as the inverter losses would be avoided. Also the system extension in blocks was found to be possible. The second study was about inverter stacking where multiple inverters can work as one unit. This type of connection allows only the required number of inverters to run while shutting down the excess ones. This would allow the converter-unit to run with higher efficiency and lower power consumption could be achieved. In future with higher loads the system could be easily extendable by connecting more inverters either in parallel or series depending on what is needed. Multiple inverters would also offer higher reliability than using one centralized inverter. The third study examined how the choice of location for a centralized power generation affects the cable sizing for the system. As a result it was found that centralized power generation should be located close to high loads in order to avoid long runs of thick cables. Future loads should also be considered when choosing the location. For the educational center the potential locations for centralized power generation were found out to be close to the school buildings and close to the dormitories.

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1 Introduction

Non-Governmental Organization Mavuno Project is building an educational center in Chonyonyo, Tanzania. The educational center will provide secondary school and dormitories to young women and education in agriculture and entrepreneurship skills. (Bahati, 2012.) This thesis is done for Engineers without borders in Sweden and its aim is to work as a suggestion for the future electrification of the educational center.

1.1 Chonyonyo, Tanzania

Chonyonyo is located in the Karagwe district in the north western corner of Tanzania. The geographical coordinates for Chonyonyo are 1° 34' 13" South, 31° 3' 55" East. Karagwe is a rural district with close to one million people. Most of the inhabitants in Karagwe are smallholder farmers. The landscape is hilly and the settlements, like the educational center, are mainly located in the ridges to avoid malaria. (Bahati, 2010.) Like many rural areas in Africa, Chonyonyo does not have access to a power grid.

Tanzania is located in Eastern Africa bordering the Indian Ocean. Tanzania's total area is 947.000 km² making it world's 31st largest country. The population is estimated at 48.26 million, which makes Tanzania world's 27th most populated country. Total electricity consumption was estimated in 2009 to be 3.589 billion kWh and in this category Tanzania is the 128th largest in the world. (Central Intelligence Agency, 2013.)

The climate in Tanzania varies from tropical along the coast to temperate in highlands. The highest point in Tanzania and Africa is Kilimanjaro at 5895 m. Current issues with the environment include: soil degradation, deforestation, desertification and destruction of coral reefs. (Central Intelligence Agency, 2013.)

1.2 Energy Poverty

The lack of access to modern energy services is a problem in Sub-Saharan Africa. According to International Energy Agency's study on electricity access in 2009 the rural electrification rate in Sub-Saharan Africa was only 14.2%, the electrification rate of Tanzania was 13.9% and 37.7 million people in Tanzania lacked the access to electricity (International Energy Agency, 2011). In the figure 1.1 the electrification rate (percentage of households with an electricity connection) and population without electricity is shown in Sub-Saharan Africa. The figure 1.2 shows how the level of access to the electricity in rural areas has progressed in the past.

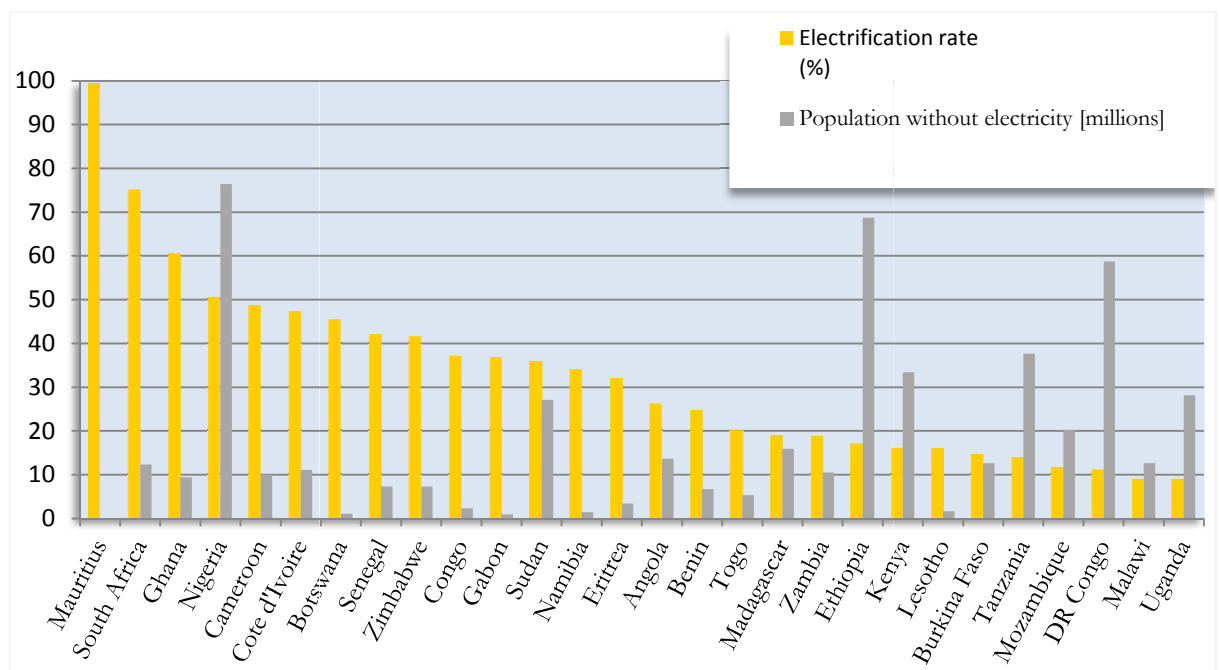


Figure 1.1: Electricity access in Sub-Saharan Africa in year 2009(International Energy Agency 2011).

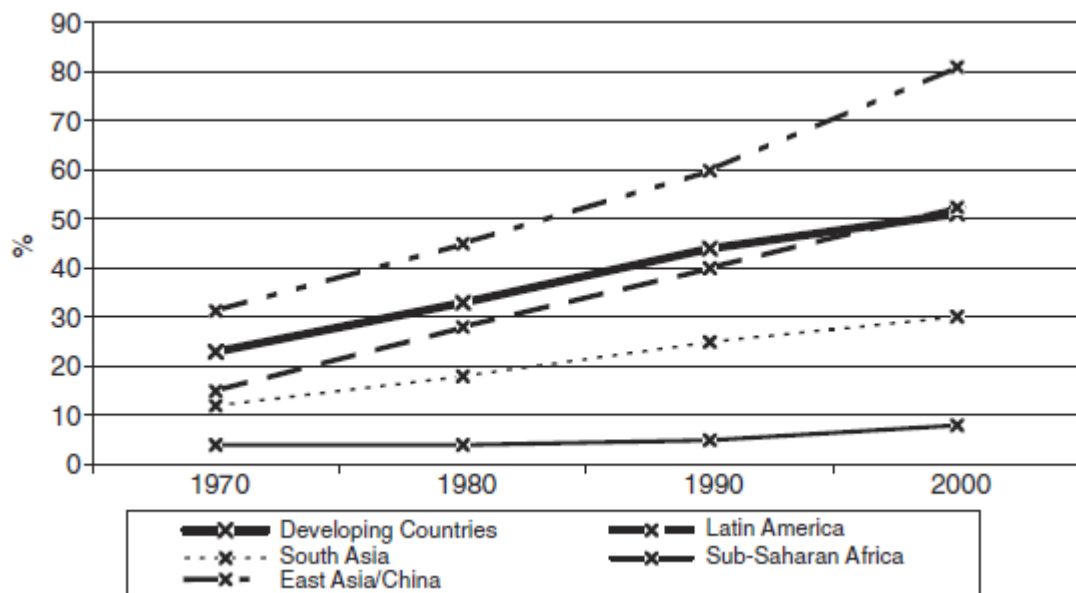


Figure 1.2: Access to electricity in rural areas (Haanyika 2005).

Energy poverty is a hindrance for economic growth. The lack of access to electricity is reality for 1.6 billion people globally and it restrains public services and industry, for which electricity is essential, from developing. In Sub-Saharan Africa 650 million people and world widely more than 2.2 billion people rely on traditional use of biomass as their primary cooking fuel. (International Energy Agency, 2010.) Gathering the fuel and cooking is traditionally a job for the women and children in the Sub-Saharan countries. World Health Organization estimates that cooking and heating with open fires and leaky stoves causes annually almost 1.6 million premature deaths from diseases, such as lung cancer and asthma, caused by indoor air pollution. (World Health Organization, 2010.) Other major problems linked with traditional use of biomass as primary fuel for cooking are that the gathering of fuel can cause deforestation and it takes excessive amount of time for the people and limits their other activities, for instance education.

Fatih Birol (2007) estimates that world population growth will increase the number of people relying on traditional use of biomass to 2.7 billion despite economic and

technological progress in developing countries. Fatih Birol also states that “The first of United Nations’ Millenium Development goal to eradicate extreme poverty is very unlikely to be met.” and that the world can’t just wait for the developing countries to afford modern energy services, instead he urges us to take action and help to accelerate energy development in poor countries. As he puts it: “Rich industrialized countries have an important role to play in this process, in addition to moral issues involved, we have obvious long-term economic, political and energy-security interests in helping developing countries along the path to energy development. For as long as poverty, hunger and disease persist. The poorest regions will remain vulnerable to humanitarian disasters, to social injustice and to political instability. Lack of resources is not an excuse. The cost of providing assistance to poor countries may turn out to be far less than that of dealing with the instability and insecurity that poverty creates.” (Birol, 2007.)

1.3 Rural Electrification

The costs for extending the grid and supplying rural areas are high. Most rural communities are characterized by low population density, poor households and relatively low consumption. The combination of these factors makes supplying rural areas expensive. (World Bank Group, 2010.)

From figure 1.3 it can be seen that fossil fuels are the major primary energy types for power generation in Africa. However South Africa alone is responsible for almost all of the coal consumption for power generation. Mainly the North African countries are responsible for the use of natural gas and also largely on the use of oil for power generation.

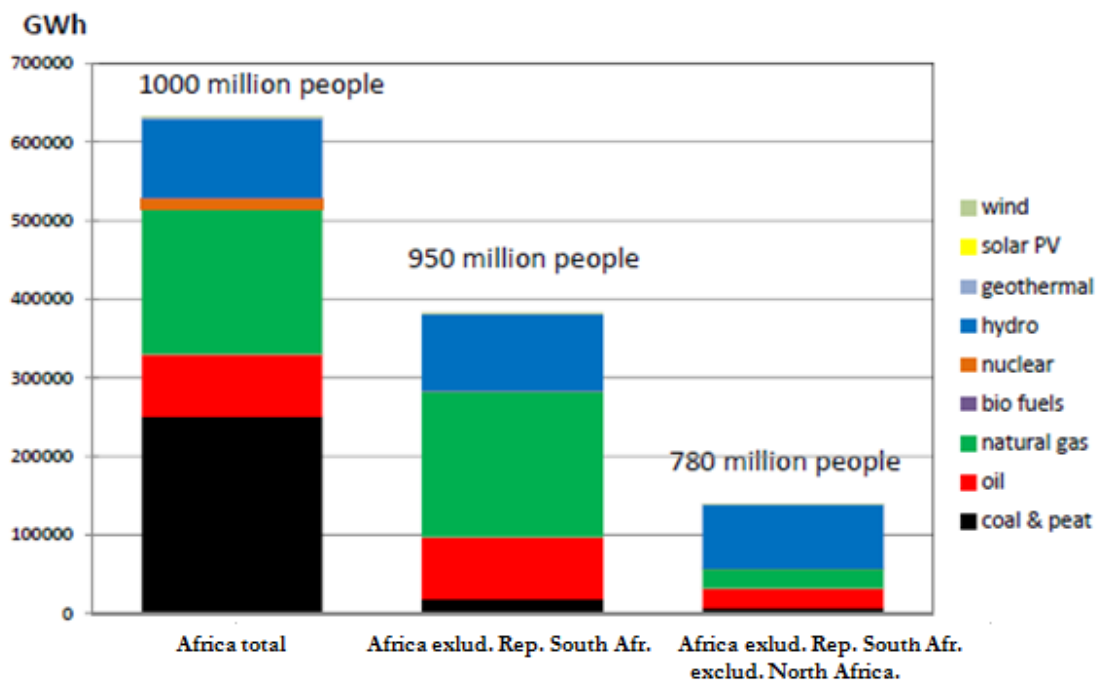


Figure 1.3: Primary energy type for power production in African areas (Klimstra 2012).

“Africa has great potential for renewable energy, especially for solar PV systems, hydro power and geothermal sources” (Klimstra, 2012). For rural electrification distributed PV generation could be a good solution. The problem with PV generation is the high investment costs which are keeping people in Africa from investing in it (Klimstra, 2012).

Electrification is essential for the development in rural areas. Electricity is needed in health care, clean water acquisition, mechanical work, and it would allow industrializing and create jobs. The lack of jobs is also a problem for Chonyonyo and Karagwe, where no

industrial development has taken place. As there isn't much of alternatives to being smallholder farmer, educated people tend to leave the region. (Bahati, 2010.)

1.4 Traditional gender roles

The Tanzanian society is largely patriarchal and especially in the rural areas traditional gender roles are strongly present (the Government of the United Republic of Tanzania www-site). Domestic duties, pregnancy and early marriages are distracting girls from finishing their schooling (Bahati, 2012). In Tanzania, on university level, only 17% of the total enrolment are girls (the Government of the United Republic of Tanzania www-site).

“When women have equal access to education, and go on to participate fully in business and economic decision-making, they are a key driving force against poverty. Women with equal rights are better educated, healthier, and have greater access to land, jobs and financial resources.” (United Nations Development Programme, 2013.) The educational center and its boarding school for girls will make an impact and contribute to the gender equality in the region.

1.5 Mavuno Project

Mavuno Project is an organization operating in Karagwe. “Mavuno programs are oriented towards advancing the development of rural Tanzanians“ (Mavuno Project, 2013). “The organization was established by rural community people in 1993” (Bahati, 2013).

List of Mavuno programs (Mavuno Project, 2013programs):

- Community Secondary School, Chonyonyo
- Water tank project, picture 1.1.
- Needy Children program
- Agriculture and environmental conservation
- HIV/AIDS Project
- Biogas program
- Mavuno Microfinance Program



Picture 1.1: Educational center constructions in Chonyonyo (Ingenjörer utan gränser, 2013).

1.6 The educational center

The school in the educational center is set to start operating in year 2015. Besides secondary school education, the school will also be providing education on agriculture and entrepreneurship, aiming for sustainable farming and land use. The educational center will provide a safe educational place for the young women and accommodate them. The school will start out with 40 female students and increase the student amount to 240 by year 2021. Offering higher level of education in safe environment to the women in rural area is a mean to break the cycle of women getting alienated from power structures due their lack of education caused by traditional gender roles distracting their studies. (Bahati, 2012.)

The educational center, seen on picture 1.2, will have dormitories for students and housing for the teachers. For agricultural means there is already storage building for crops, pilot biogas plant and in the future possibly stables and greenhouses. The biogas pilot plant uses organic matter to produce biogas. The biogas is going to be used for cooking. The educational buildings are going to be four school buildings, a library and a building for administration. The school buildings will have a laboratories and computer classrooms. The educational center will also have a canteen, a building for health care and accommodation for the staff and students. Currently the constructions of large dormitories for girls are ongoing. The smaller dormitories with small room size are almost finished. The small rooms are suitable for accommodating 16 students each. The housing for disabled students is nearly finished as well. (Bahati, 2012.)

Picture 1.2: Master plan for the Chonyonyo Education Center and Secondary school (Master plan for the Chonyonyo Education Center and Secondary school, 2012).

It is possible that the power grid will one day be extended, but as it is the educational center needs electrification. Even if the grid was to be extended to cover Chonyonyo, the quality of the grid is low and many power shortages occur from which some can be very long. The performance of laboratory equipment, computers and other sensitive electronics at the Educational Center can be affected by the poor quality power grid, therefore in this study sustainable power generation methods for electrification of the educational center will be presented.

Mavuno Project and Engineers Without Borders in Sweden have started the local electrification by installing PV panels. For one of the dorms the power system is already planned and partly installed. The educational center will be expanded over time, and the need of electricity will increase accordingly over the coming years. As for Mavuno Project and Engineers Without Borders it isn't economically possible for them to install a full power system in one go. The electrification will have to be done step by step.

1.7 Aims

In order to achieve a sustainable and economically feasible electrification for the educational center, an overall plan for the technical design of the power generation system should be followed. In this study different design suggestions will be presented with their

advantages and disadvantages. The suggestions will help the Mavuno Project and Engineers Without Borders in future in the designing of the power generation system.

1.8 Method

Simulations of the system were performed with the simulation software HOMER to get an approximately optimized system with component sizes for a specified load. The optimization process was done step by step from large range of values to gradually smaller value range to find out the best system. HOMER is a tool which bases the optimization on economic factors, which is to say that changes in costs play a major role in the optimization process. Precise inputs with Homer are required for accurate results.

The system looked at was a PV/diesel generator hybrid system. This type of a system was chosen because Chonyonyo has plenty of solar radiation round the year due its close location to equator. The diesel generator is included in the system to provide backup. Other renewable energy sources were not considered as the location doesn't have resources for hydro power and the wind speeds are quite low. After optimizing the base case system with HOMER, further optimization possibilities were studied.

The project tasks:

- Literature study
- Acquiring of weather data, resource data and component data
- Creation of load profile from given estimation
- Optimization of the systems with simulation software HOMER
- Further optimization possibility study

1.9 Design Software HOMER

The system optimization and simulation was done with the Hybrid Optimization Model for Electric Renewables (HOMER) design software developed by the National Renewable Energy Laboratory. Homer can be used for the designing and analyzing of hybrid power systems. The simulations can be done for grid-tied and off-grid environments. The simulated system can consist of several different types of power generators, energy storages and loads.

1.10 Previous Studies

Givler & Lilienthal (2005) found out in their study that “the PV/battery systems were most cost-effective up to loads ranging from 3 kWh/d to 13 kWh/d depending on the reliability, solar resource and diesel fuel price”. And for loads above that PV/Hybrid systems were the best choice. They also found out that accepting some capacity shortage factor improves the competitiveness of PV/battery systems. The improved competitiveness is based on the lower capital costs, as pursuing high availability with a PV/battery system is very expensive. (Givler & Lilienthal, 2005.)

Rohit Sen (2011) performed in his studies simulations with HOMER on an off-grid electricity generation with renewable energy technologies and compared an optimized hybrid system to an expensive grid extension. His conclusion was that decentralized renewable energy technology off-grid system is the best alternative to grid extension and

can be cost effective even if the grid connection was possible. He also states that system can't provide a viable solution if it only relies on one energy source (Sen, 2011).

Mondal & Denich (2010) conducted a simulation study with HOMER on decentralized power generation with hybrid systems in Bangladesh. The aim of the study was to find optimized hybrid system to fulfill 50 kWh/day primary load and 11 kW peak load. Their result was that a system with 6 kW PV array, 10 kW diesel generator and a battery bank 16 kWh was optimal for that purpose and that this kind of setups are good candidates for locations with high amounts of solar radiation. (Mondal & Denich, 2010.)

Holliman (2010) studied different system designs both off-grid and grid-connected for a rural health clinic in Nicaragua. Holliman used both HOMER and PVSYST in his study. The outcome of the study was that the best option for the health clinic was to be connected to the relatively close grid and to have a battery bank providing backup power. (Holliman 2010.)

Al-Badi, AL-Toobi, AL-Harthy, Al-Hosni & AL-Harthy (2012) conducted a study on hybrid systems for three sites, located in Oman, Masirah, Mothorah and HB Hameed. They found out that the sites were prospective candidates for hybrid systems and the payback periods for HB Hamees and Mothorah would be less than 6 years. (Al-Badi & co., 2012.)

2 Stand-Alone PV Systems

Stand-alone PV systems are not connected to the public electricity grid. There are three main categories of stand-alone PV systems: systems providing DC power only, systems providing AC power through an inverter and hybrid systems (diesel, wind, and hydro). (The German Energy Society, 2008, 297.) In figure 2.1 there are examples of stand-alone PV/hybrid systems.

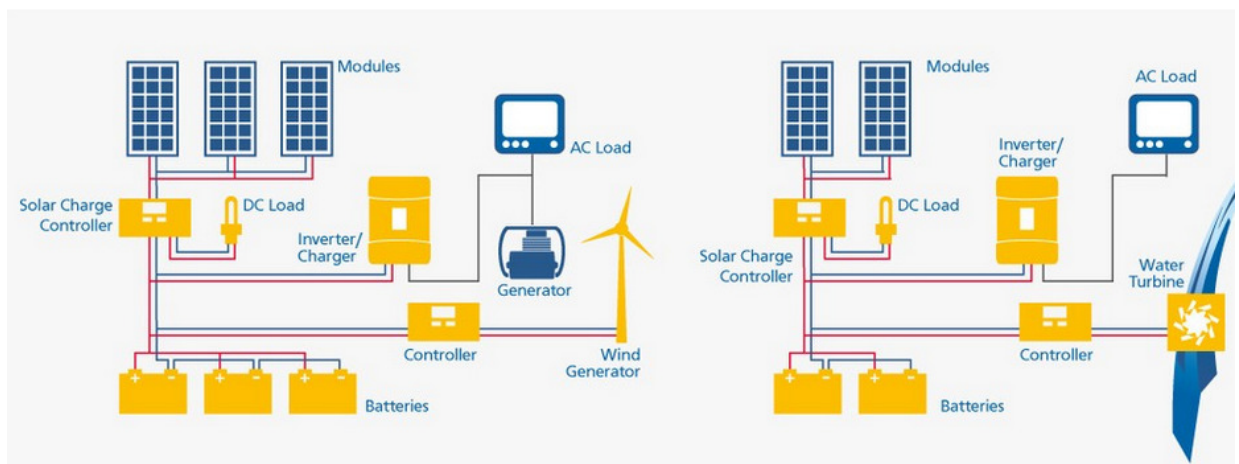


Figure 2.1: Typical PV/hybrid stand alone systems (Phaesun 2013).

2.1 Stand-alone PV/Hybrid system components

PV Stand-alone system consists of many components. In a PV stand-alone DC system the needed components are usually the PV modules, charge controller and the battery bank. In AC systems an inverter is required for converting the DC to AC. Often larger stand-alone systems have more than one way of producing electricity to achieve higher availability and

reliability for the system. The PV electricity production can be accompanied with renewable wind or hydro power or with a diesel generator.

To ensure high availability diesel generator can be a good choice. Diesel generator requires relatively low initial investments compared to using only PV and batteries as the battery size can be reduced without compromising the system availability (Markvart, 2000, 120). With a diesel generator, one should also notice that it isn't as maintenance free as PV and it will cause pollution and noise.

2.1.1. PV power generators

PV generator consists of solar cells that can convert the solar radiation into DC electricity through photovoltaic effect. Most commonly solar cells are made from semiconductors but photovoltaic materials are not restricted to them. There are also solar cells which utilize organic molecules in the conversion of sunlight to electricity (Markvart, 2000, 25-26.) The most common solar cell type is crystalline silicon. From which there are two main types, monocrystalline silicon cell and multicrystalline silicon cell. Other common solar cell types are thin-film cell, amorphous silicon cell, copper indium gallium diselenide (CIGS) and cadmium telluride.

2.1.2. Diesel generator

“Diesel generators used in hybrid energy systems are synchronous alternators, which are directly coupled to a diesel engine (Markvart, 2000, 127)”. The engine is controlled by adjusting the fuel flow. Frequent starting causes wear on the engine and the engine should run for minimum time period to reduce wear. Running the engine in low, less than 40% of rated capacity, power level should be avoided as the engine will run with low efficiency and it can potentially cause glazing on the cylinder walls. Typical diesel engine's optimum fuel efficiency can be achieved by running the engine above 80% of its rated capacity. (Markvart, 2000, 127-128.)

2.1.3. Energy Storage

In general stand-alone systems need to have energy storage to achieve higher availability or to be able to supply higher power at times. The most commonly used battery type in stand-alone systems is a rechargeable lead-acid battery. They are cost-effective and work efficiently with a variety of charging currents (The German Energy Society, 2008, 298).

2.1.4. Charge controller

A charge controller is needed to prevent excessive discharging or overcharging of batteries. Deep-discharging is avoided by monitoring the battery voltage and having the load disconnect from it when a minimum preset value for the voltage is met. The load is allowed to connect to the battery once the voltage has risen sufficiently. The overcharging can be prevented with either shunt or series regulator. (Markvart, 2000, 100-101.) The charge controller is also able to keep the batteries from discharging during low irradiances with a blocking diode preventing the reverse current caused by the plummeting PV voltage (The German Energy Society, 2008, 301).

2.1.5. Converter

An inverter is needed in a stand-alone system to convert the DC generation into AC in order to be able to use conventional 230 VAC appliances. A rectifier is used for converting the AC to DC, for instance when AC power generator like a diesel generator is needed for recharging the batteries.

There are three types of inverters on the basis of what type of alternative current they are able to produce. Sine-wave inverters produce high quality AC which works even with sensitive electronics. Modified sine-wave and square-wave inverters are cheaper than sine-wave inverters and they don't produce high quality sine-wave AC and shouldn't be used with sensitive equipment that could be damaged. (The German Energy Society, 2008, 313-315.)

3 System design

This chapter covers an optimization simulation of a power generation system for the educational center. The simulation is done with the design software Homer. This was done in order to find out the approximate size for the power generation system.

For the simulation load estimations which are found in the Appendix A, component data, resource data and weather data were input to the software. In addition to these inputs HOMER has many other options that can be adjusted to get more specific system.

3.1 Weather Data

For the weather data, both the temperature and solar radiation data from NASA database (<http://eosweb.larc.nasa.gov>), were chosen to be used for the simulations. In figures 3.1 and 3.2 the acquired data has been input in HOMER. Clearness index is a measure of the clearness of the atmosphere and its typical values range between 0.25 and 0.75 from which the latter would be a very sunny month (Lambert, 2004).

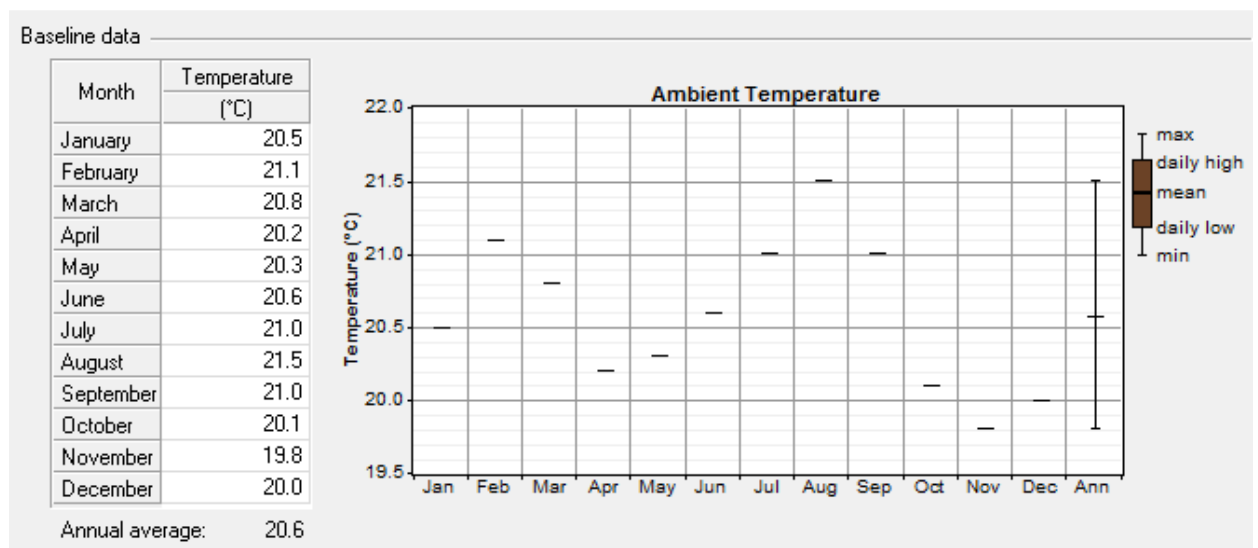


Figure 3.1: Temperature data for Chonyonyo.

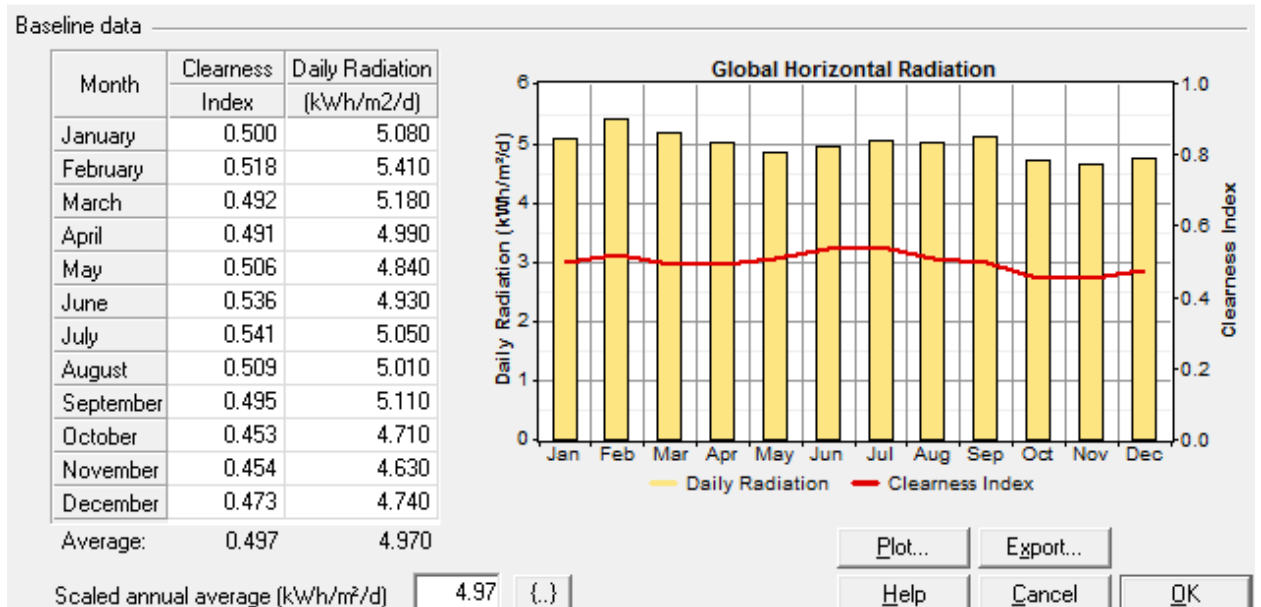


Figure 3.2: Solar radiation data for Chonyonyo.

3.2 Load estimations

The loads used for simulation are based on estimations made in year 2011. They have been input to HOMER as can be seen in the Figure 3.3. The load estimations can be found in the Appendix A.

The day-to-day and time step variability were both set to 15% to allow some variability in the loads. By default HOMER suggested 15% for day-to-day and 20% for the time step, but the latter was changed as it is assumed that the community is aware of their consumption and plans it to some extent and try to avoid unnecessary very large peak loads. The scaled annual average consumption is 74.7 kWh/day.

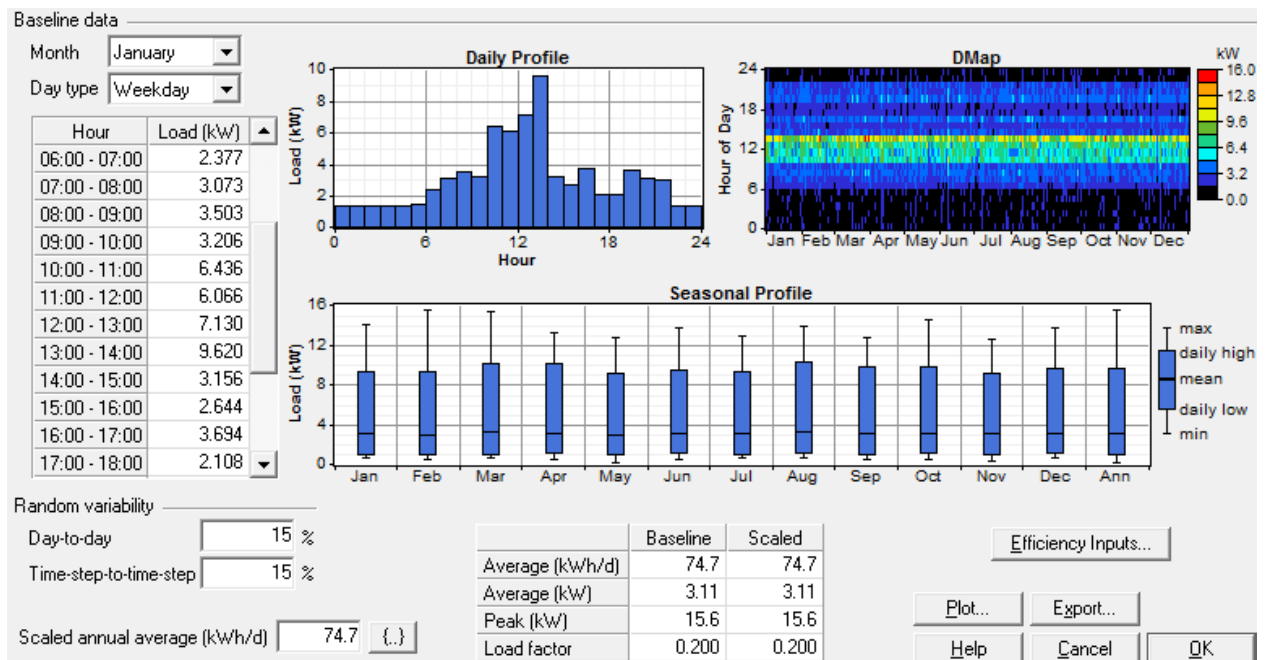


Figure 3.3: Load profile for the educational center.

3.3 System Components

The equipment, the design software HOMER considered in the simulation, is shown in the figure 3.4. The equipment consists of following components: a diesel generator, PV array, battery bank and the converter. For the sheer simplicity of the system it was decided that only one diesel generator was simulated. Besides solar energy no other renewable energy sources were considered for power generation. The specific components were chosen so that they would represent well the average costs and the information, such as efficiency, required by the simulations would also be common. Although charge controller is vital for the system it doesn't contribute to the simulations done with Homer.

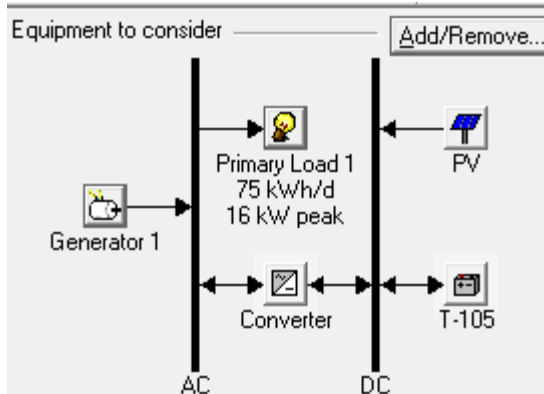


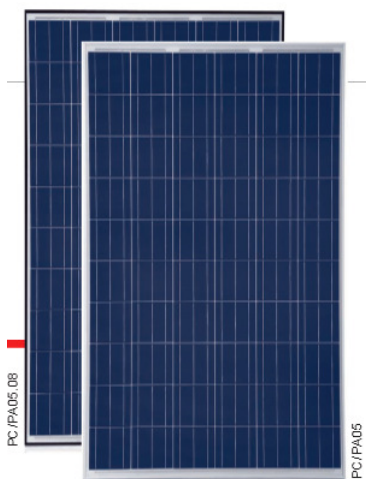
Figure 3.4: Equipment simulated for the system optimization in HOMER.

3.3.1. PV modules

The multicrystalline Trina Solar 230 W PV module, as seen on the picture 3.1, was used in the simulation. The values from the module HOMER needed for the simulations were:

- Temperature coeff. of power $-0.43\% / ^\circ\text{C}$
- Nominal operating cell temp. $46\text{ }^\circ\text{C}$
- Efficiency at std. test conditions 14.1%

Ground reflectance was set to 25% which would be a typical value for a fairly green area. Other input parameters for PV were left at their default values.



Picture 3.1: Trina Solar 230 W PV module (Trina Solar, 2012).

3.3.2. Batteries

For the system a battery bank is needed to provide the electricity when the power generation for any reason isn't matching the energy consumption. The battery type that was chosen for the simulations is Trojan T-105P 6 V 225 Ah. The batteries are in strings of eight with fourteen strings parallel.

12 VDC and 24 VDC systems were ruled out for their high currents and high conversion losses. The system voltage could also have been higher for instance 110 VDC but the 48 VDC was chosen over it, for in the simulations 48 VDC resulted in lower costs. Also many inverter/chargers available for off-grid systems work in these slightly lower DC voltages. The battery is on the picture 3.2.



Picture 3.2: Trojan T-105 battery (Trojan Battery Company, 2013).

3.3.3. Converter

For the system simulation 48V DC 5 kW Victron MultiPlus inverter/chargers were chosen. Three MultiPlus units can be configured for three phase output and six of these sets can be parallel connected totaling to 72 kW inverter power. If connected to one phase up to six Victron units can be installed parallel. The Victron MultiPlus can automatically control the diesel generator while maximizing solar power usage. (Victron Energy, 2013a) Picture 3.3 demonstrates the looks of the unit. The input parameters for the converter were left at their default values as they matched with the MultiPlus inverter/charger.



MultiPlus
24/3000/70

Picture 3.3: Victron MultiPlus 3 kW inverter (Victron Energy, 2013).

3.3.4. Diesel generator

The diesel generator chosen for the simulation has a rated 5.5 kW maximum power and can be seen on picture 3.4. It is portable and has a soundproof enclosure.

At first simulations were done with larger range of diesel generator sizes and the optimized PV/hybrid system had a small 1 kW diesel generator. In comparison to 5.5 kW diesel generator the small generator would run more often and usually at full power. Also the small diesel generator resulted in slightly smaller system size and costs, but the availability was only around 96%. However a larger one was chosen because the loads will be higher in the future and a small 1 kW diesel generator wouldn't much of a help if the PV generation was down for a longer period of time.



Picture 3.4: Automatic 5,5 kW Diesel generator (Central Maine Diesel, 2013).

3.3.5. Component costs

The costs for PV modules, batteries and inverters were taken from a South African online store “Sustainable.co.za”. However the diesel generator’s information is from an American seller. The price for diesel in Tanzania is around 0.94 €/l (IOL, 2012). The component costs are listed in Table 3.1. For the system, no installation maintenance or design costs were considered.

Table 3.1: Component prices used in the simulations.

Component	Name	Cost [€]
Inverter 3 pcs ¹	Victron : MultiPlus 5 000 W	7 800
PV module 11 pcs ²	Trina Solar : TSM-PC05 230 W	2 840
PV module 25 pcs ³	Trina Solar : TSM-PC05 230 W	6 250
Battery ⁴	Trojan T-105	165
Diesel generator ⁵	5.5 kW	1 900

¹ (Sustainable.co.za, 2013), ^{2,3} (Sustainable.co.za, 2013a), ⁴ (Sustainable.co.za, 2013b),

⁵ (Central Maine Diesel, 2013)

3.4 Sensitivity and lifetime inputs for the simulation

Instead of the default 60 minute time step, the simulations were done in five minute steps to allow HOMER to optimize the system more precisely. Imported data is down-sampled and up-sampled by HOMER to match the chosen time step. A maximum capacity shortage of 5% was allowed. The 5% was chosen as it was assumed this level of shortage

would be still acceptable and raise PV generations competitiveness. The lifetime set for the converter was 15 years and for PV modules 25 years.

The diesel generator was set to cycle charging which means that while running, the diesel generator would be running in full power to charge the batteries. Also a setpoint for state of charge was applied. This means the diesel generator will stop charging the battery bank once the setpoint is reached. In the table 3.2 can be seen how different State of charge setpoints affected the system. The 80% setpoint is the one used in the further system optimizations for it resulted in lower costs and kept the state of charge of the batteries high enough to ensure a long life time for them.

Table 3.2: The affect of state of charge setpoint on costs.

State of charge setpoint	PV [kW]	T-105	Conv. [kW]	Initial Capital [euro]	Total NPC	COE [€/kWh]	Diesel [L]
80%	26	112	15	55 875	92 121	0,264	1 154
90%	27	112	15	56 934	94 660	0,272	1 322
100%	20	8	10	29 561	145 310	0,417	8 290

3.5 Simulation results

The optimization process was started with large size range for each component. The range was gradually narrowed to get smaller steps between component sizes to find out the best system. The narrowing of component size range was done in order to keep the simulation times sensible. In the end of this process results in three different categories were achieved. The system categories are based on what equipment was used. The results are shown in figure 3.5. The reason there is no category with only diesel generator is that the very large ones were dropped out earlier in the iterative simulation process for these systems had very high total net present cost. The net present cost tells us the difference between the present value of all the costs that incur in the system's life time and the present value of its lifetime revenue.

The system without any PV modules relying only on diesel generator and batteries has very low initial costs but in the long run it will be almost twice as expensive as the two other optimized systems. Worth of mentioning is also that the simulation was done with fixed fuel costs and that the fuel costs can rise in the future.

The system with no diesel generator has the lowest levelized cost of electricity. However the lack of backup generation can be a problem. The inverter can be seen as the weak point in a PV system and in a case of failure this system would be troublesome. The third option, the hybrid system, with both PV modules and diesel generator is the most versatile and the most reliable of them while providing the lowest capacity shortage and almost as low levelized cost of electricity as the PV only system.

The hybrid system consists of 26 kWp of PV; 5.5 kW diesel generator, three Victron inverters with a combined 15 kW power and a battery bank of 112 Trojan T-105 batteries, with a total capacity of 151 kWh providing autonomy of 34 hours for the system. In figure 3.6 it can be seen that the battery costs dominate the systems net present cost.

In the results the operating costs are the sum of the fuel costs and the component replacement costs minus the component salvage values, which are calculated by HOMER.

The calculated salvage value of a component is directly proportional to its remaining life. No maintenance costs are considered. (Lambert, 2004).

		PV (kW)	Label (kW)	T-105	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage	Diesel (L)	Label (hrs)
		27		152	15	\$ 61,634	1,686	\$ 83,186	0.243	1.00	0.05		
		26	5.5	112	15	\$ 55,875	2,835	\$ 92,121	0.264	0.91	0.00	1,154	698
			5.5	16	10	\$ 9,540	11,641	\$ 158,352	0.463	0.00	0.05	10,356	6,962

Figure 3.5: The Simulation results: Optimized systems by different categories.

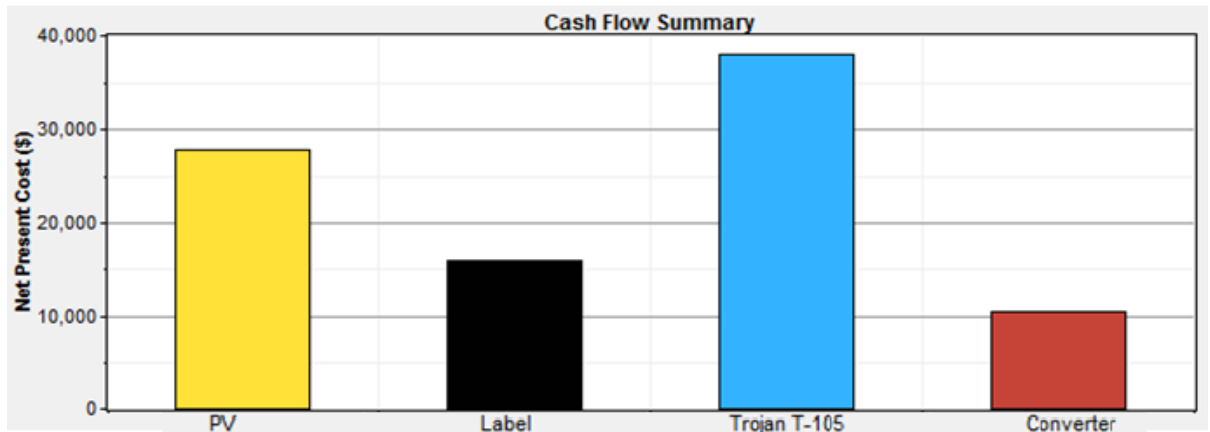


Figure 3.6: The Simulation results: Cash flow summary of the PV/hybrid system

3.5.1. Hybrid System: PV array

The rated PV array capacity for the hybrid system is 26 kW. In the figure 3.7 the power output is shown during a year. The mean output of the PV array is 4.1 kW and the maximum output 22.7 kW. The yearly production is almost 36 MWh. The daily PV array power output profiles can be seen on the figure 3.8. From the profiles one can see the power generation starting early at six o'clock lasting for 12 hours and peaking around 12 o'clock midday. The daily average peak power output is relatively stable over the year varying between 12 kW in November and 15 kW in February. PV penetration for this system is 132%. PV penetration means the PV power output divided by the mean primary load.

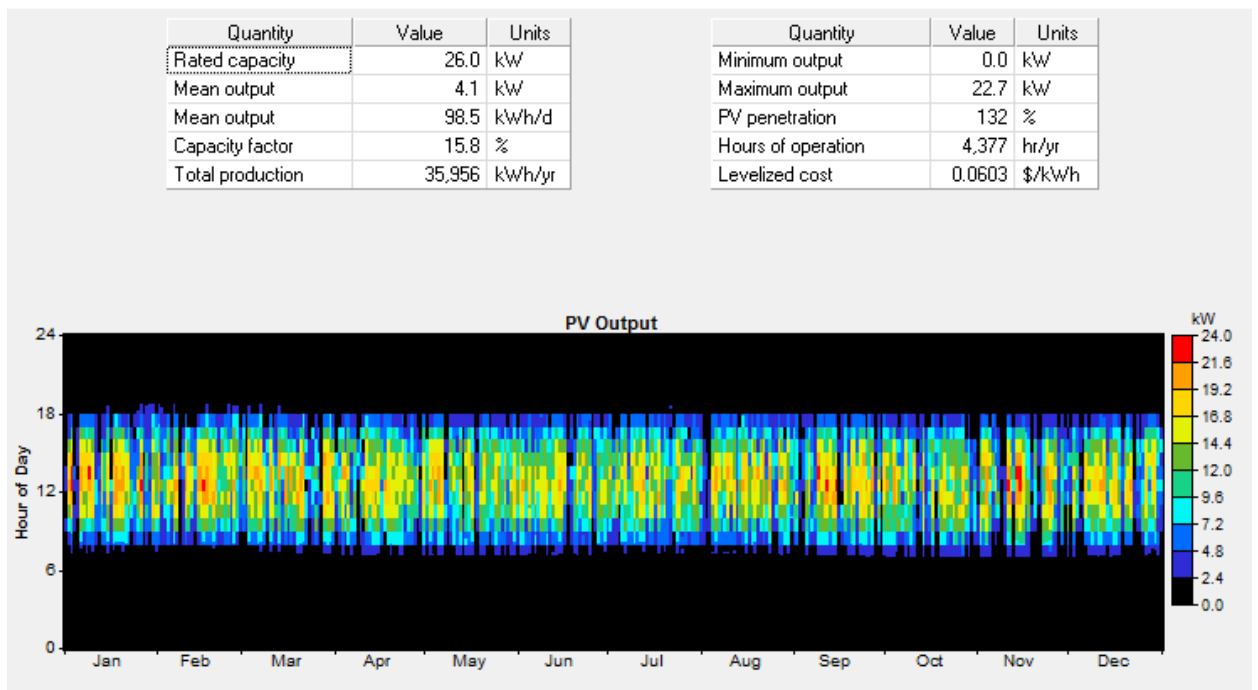


Figure 3.7: The Simulation results: PV array information.

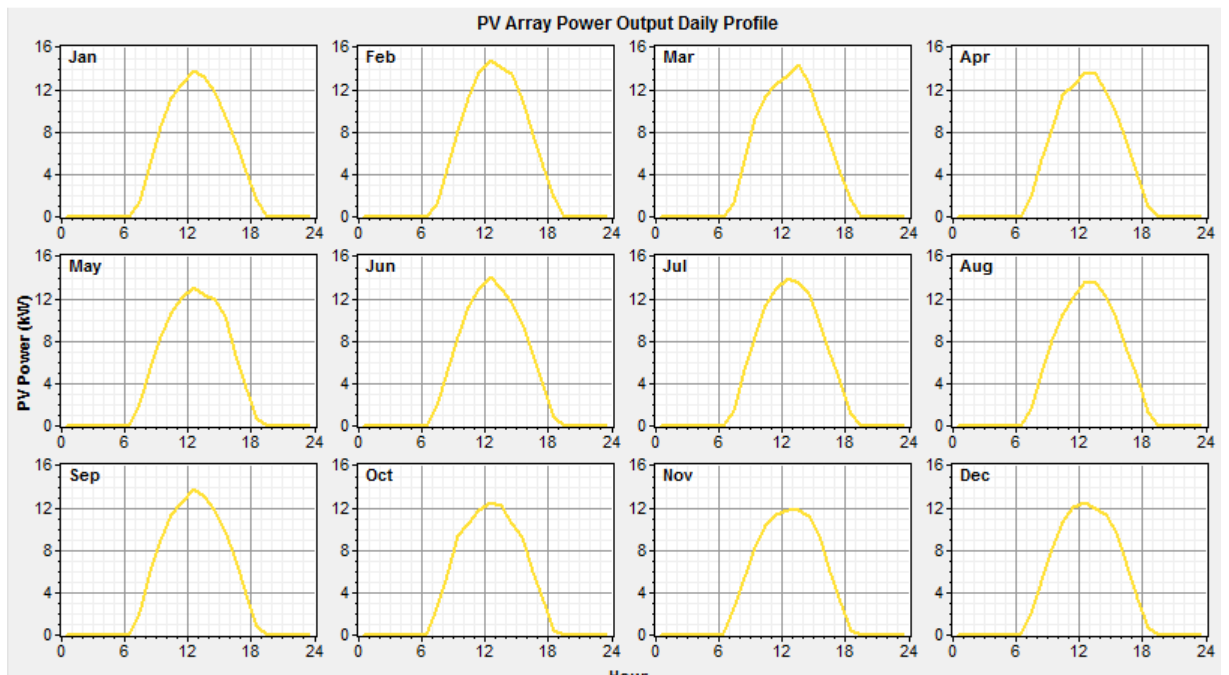


Figure 3.8: The daily power output profile for the PV array.

3.5.2. Hybrid System: Converter

For the optimized hybrid system a 15 kW (3 x 5 kW) converter was chosen.

The figure 3.9 shows system's inverter and rectifier power outputs during the year. It needs to be noted that instead of simulating all the possible converters individually HOMER treats them as one block.

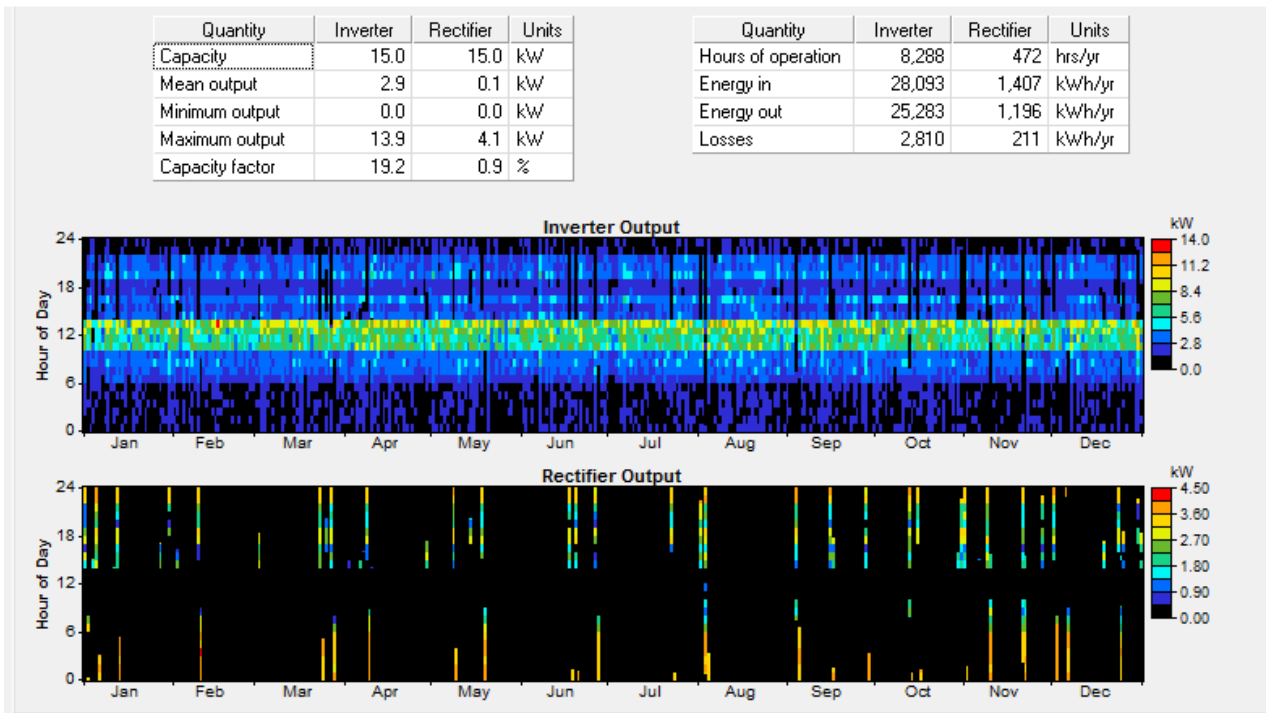


Figure 3.9: The Simulation results: Converter information.

3.5.3. Hybrid System: Battery bank

The simulation results for the hybrid system in figure 3.11 gives an estimated lifetime of 8 years for batteries. The autonomy time achieved with the system is 34 hours. From the aforementioned figure it can also be seen that the state of charge of the battery bank is above 80% for most of time and it is not dropping too often below 50%, which is important for gaining a long service life (The German Energy Society, 2008, 325). The relevancy between depth of discharge and the number of cycles for the chosen battery can be seen in the figure 3.10. The figure 3.12 shows the input power for the batteries.

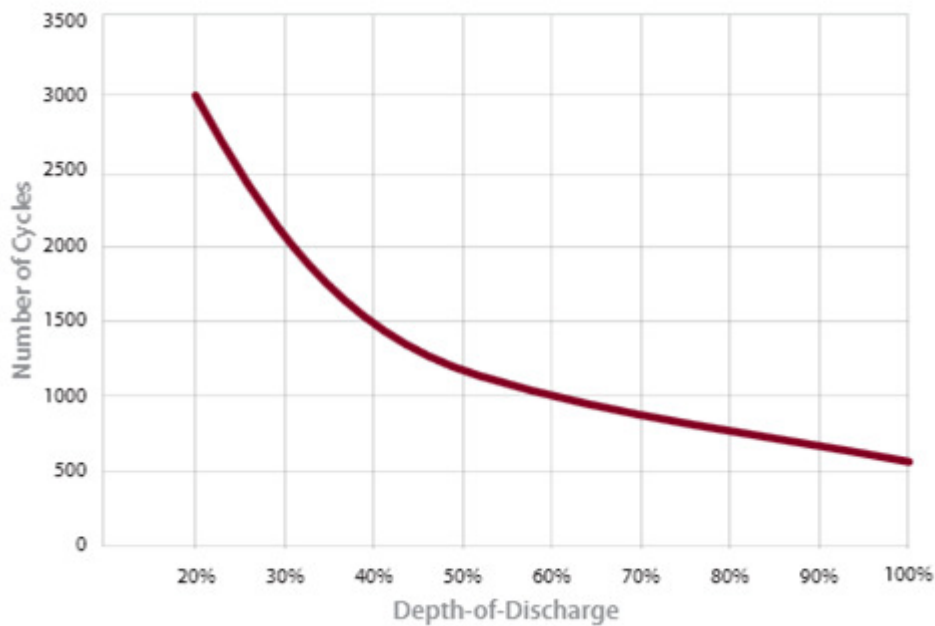


Figure 3.10: Relevancy between depth of discharge and number of cycles on Trojan T-105 (Trojan Battery Company, 2013).

Quantity	Value
String size	8
Strings in parallel	14
Batteries	112
Bus voltage (V)	48

Quantity	Value	Units
Nominal capacity	151	kWh
Usable nominal capacity	106	kWh
Autonomy	34.0	hr
Lifetime throughput	94,640	kWh
Battery wear cost	0.212	\$/kWh
Average energy cost	0.040	\$/kWh

Quantity	Value	Units
Energy in	12,735	kWh/yr
Energy out	10,869	kWh/yr
Storage depletion	47	kWh/yr
Losses	1,819	kWh/yr
Annual throughput	11,789	kWh/yr
Expected life	8.03	yr

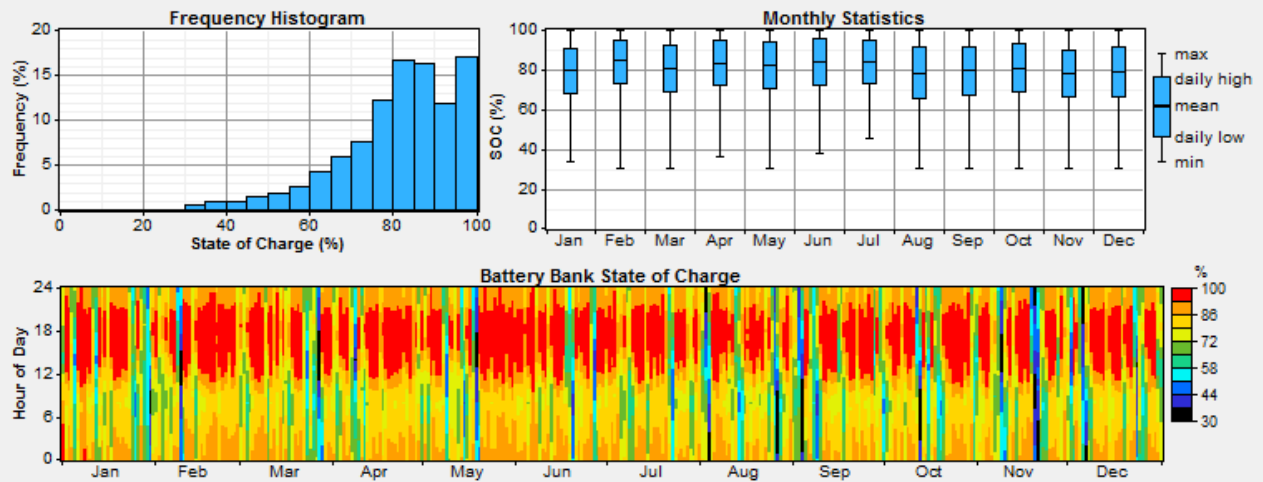


Figure 3.11: The simulation results: Battery information.

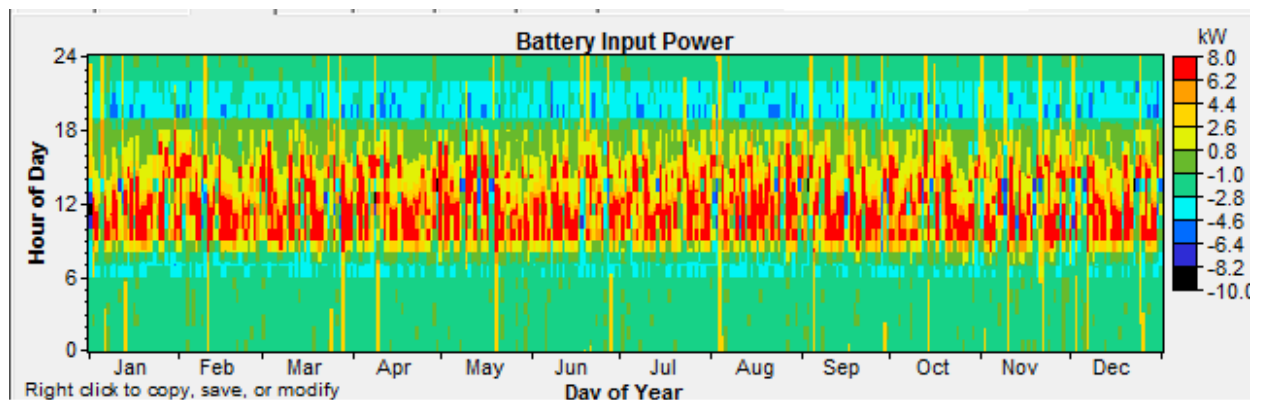


Figure 3.12: The simulation results: Battery input power.

3.5.4. Hybrid System: Diesel generator

In the optimized hybrid system for most of the time the diesel generator is operating close to its rated power as can be seen on the figure 3.13. This is good as engines tend to run with higher efficiency close to their rated power. The diesel generator is operating annually nearly 700 hours. While operating the mean electrical output of the diesel generator is 4.85 kW and the minimum electrical output is 1.65 kW. The annual fuel consumption is roughly 1150 liters.

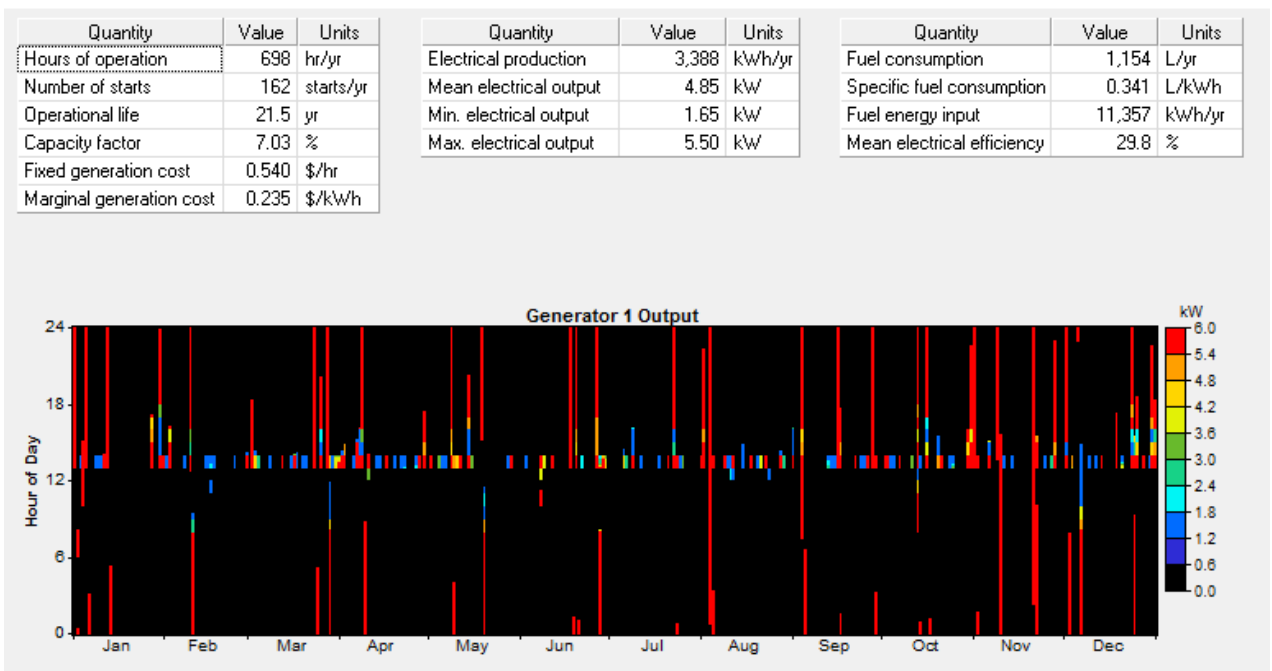


Figure 3.13: The simulation results diesel generator information.

3.5.5. Hybrid System: Monthly average electric production

From the figure 3.14 one can see that for the hybrid system the yearly combined total production is 39.3 MWh with a renewable fraction 0.914. From the total production only 9% is produced by the diesel generator and this keeps the total fuel costs low. The capacity shortage of the system is close to 0%. The annual amount of excess electricity production is almost 7.2 MWh.

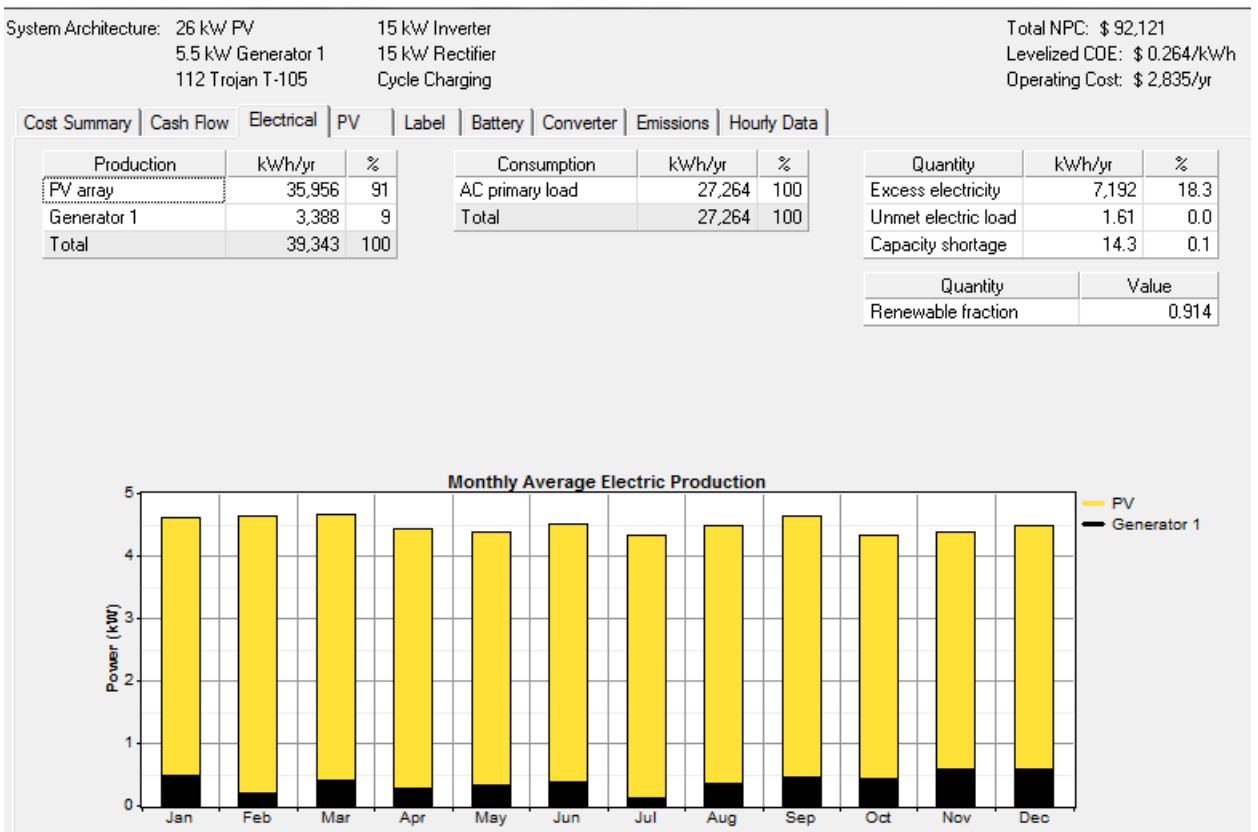


Figure 3.14: The simulation results monthly average electric production.

4 Possibilities of optimization

In this chapter, three possibilities are looked at. First the possibility of distributed power generation together with DC night loads is looked at. The second possibility reviewed is the option of connecting multiple inverters together instead of one centralized inverter. The third study investigates how centralizing the power generation at different points affects the wire sizing.

4.1 Distributed power generation with DC night loads

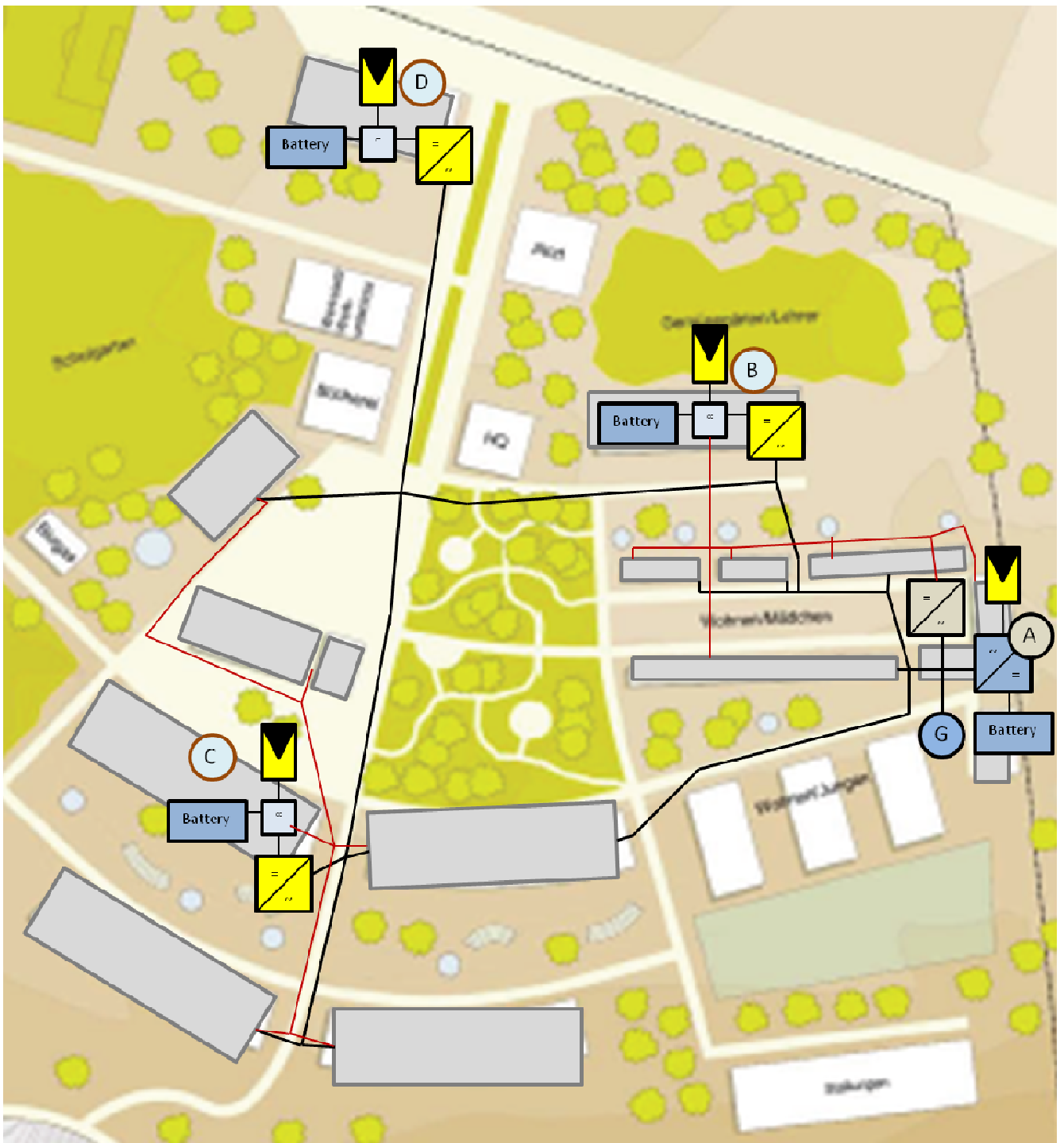
In this subchapter 4.1 two changes are made in the system design in comparison to the chapter 3. Firstly the power generation will be distributed to allow shorter cable lengths and lower power distribution losses. Secondly there will be DC night loads in order to achieve lower power consumption at night.

4.1.1. Introduction

Picture 4.1 shows the possible locations considered in subchapter 4.1 for distributed power generation. The inverter, possibly Victron MultiPlus which was also used for the system in chapter 3, in unit A located close to the House for disabled could be the one forming the grid. The PV power generation is distributed close to DC night loads and future buildings in order to have short cable runs. Power generation units able to produce DC for the night loads have their PV array, charge controller, battery bank with chosen VDC and a bi-directional inverter. Examples of bi-directional inverters that could connect the power generation units to the AC grid are for instance Victron Multiplus or Quattro inverter/chargers. In future, if the system needs to be extended, more units can be added to the AC grid via bi-directional inverters. Extending the system with just a PV generation unit can be done via grid inverter such as Victron BlueSolar grid inverter.

The highest daily peak loads are at the school buildings and storage facilities, thus these locations should be considered for the PV power generation. In this system the security lights are considered as the DC night load.

The PV generation at the unit-D could in the future provide DC also for the closely situated doctor house, workshop and library. Unit-B could in the beginning provide DC to the teacher's housing, dormitories, house for disabled and the toilets and in the future for the HQ as well. The diesel generator could be connected to the DC grid. But since the DC loads are relatively low as can be seen from the table 4.1 it might not be worthwhile. The unit-C located at the school buildings will provide DC also for the dining hall and administration.



Picture 4.1: PV power generation units **A**, **B**, **C** and **D** distributed closer to day load with the possibility to run DC night loads.

Table 4.1: Building load and consumption.

Building/Load	Peak Load Max. [kW]	Total AC Consumption [kWh/day]	DC Night load (security lights) [kWh/day]
Storage facilities	4.13	10.76	0.96
Administration block	1.38	12.24	0.48
Teacher's housing	2.05	10.20	0.48
Small dormitory W	0.30	1.62	0.96
Small dormitory E	0.30	1.62	0.96
House for disabled	0.25	3.17	0.48
Toilett 2+3	0.20	1.43	0.96
Large dormitory N	0.38	1.68	0.72
Large dormitory S	0.54	2.40	1.20
Dining hall+toilet	0.88	9.10	1.68
School buildings	7.95	21.24	3.36
Total:		75.46	12.24

Notion: Toilets 2 & 3 refer to the ones close to the dormitories, toilet number 1 refers to the one next to the dining hall. To distinguish dormitories from each other a letter has been added to indicate their cardinal point in relation to each other.

4.1.2. Optimization

In Homer it is possible to simulate several loads and for this simulation there are two different loads, AC and DC which can be seen in figures 4.1 and 4.2. Other than having two load profiles and less nighttime power consumption the simulation process is done like in chapter 3. However the optimization of this AC/DC system is simplified since HOMER simulates the PV array as a one block and doesn't consider distributed generation.

The combined scaled annual average consumption and the peak load are the same as they were in the base case system. The difference is that in this system loads are divided in AC and DC loads. This however doesn't affect the AC peak load because it's during the daytime and the security lights that are now considered as DC loads, didn't contribute to that. The DC load differs from AC load in having no day-to-day or time-step variation as the security lights are expected to be switched on only on the given hours.

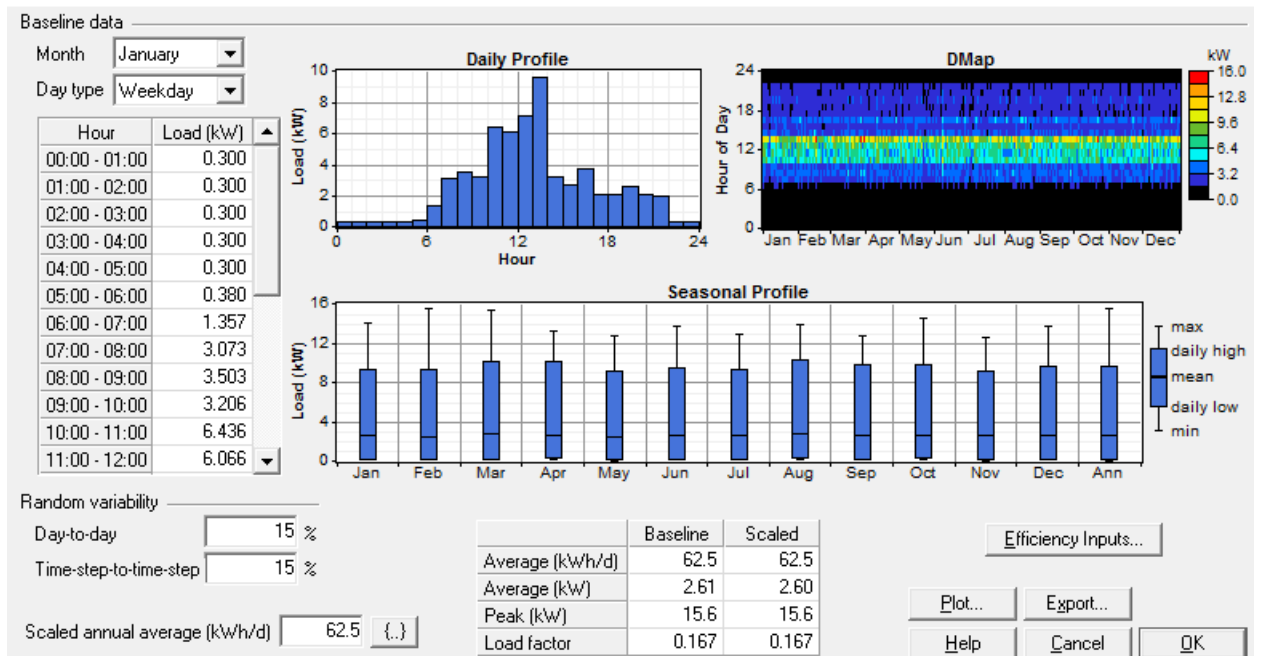


Figure 4.1: Estimated AC load profile for the system



Figure 4.2: Estimated DC load profile for the system

4.1.3. Optimized AC/DC system

In the figure 4.3 the optimized system is shown. The PV/hybrid system consists of 27 kW PV; 5.5 kW diesel generator, 88 T-105 batteries and a 15 kW converter.

The initial costs for this system are almost 3000 € lower than for the base case system and the levelized cost of electricity is 0.005 €/kWh lower. In comparison to base case, the costs are lower as the battery size has been significantly reduced. This has been possible for now the night loads are in DC and there are no necessary conversion losses. From figure 4.4 it can be seen that the cost for batteries is no longer as dominating for the total system cost as it was in the base case system (figure 3.6).

	PV (kW)	Label (kW)	T-105	Conv. (kW)	Initial Capital	Operating Cost (\$/yr)	Total NPC	COE (\$/kWh)	Ren. Frac.	Capacity Shortage	Diesel (L)	Label (hrs)
	27		144	15	\$ 60,314	1,608	\$ 80,871	0.237	1.00	0.05		
	27	5.5	88	15	\$ 52,974	2,919	\$ 90,288	0.259	0.92	0.00	1,191	750
		5.5	24	10	\$ 11,460	11,697	\$ 160,985	0.466	0.00	0.02	10,273	6,234

Figure 4.3: Optimized AC/DC hybrid system.

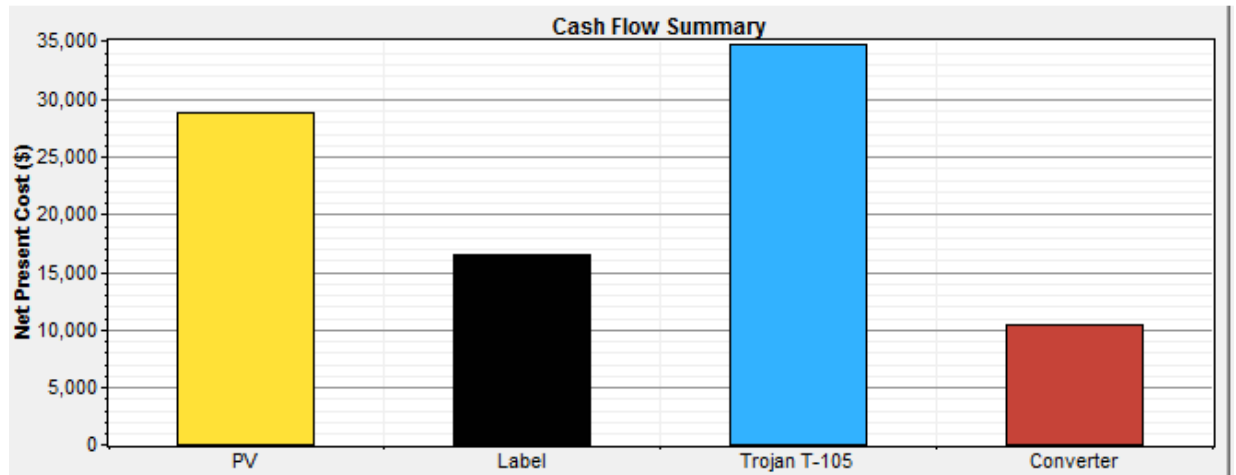


Figure 4.4: Net Present Cost for the hybrid system.

4.1.4. Electric production of the system

In comparison to the base case, In the figure 4.5 the annual production is shown to have been increased to 40.8 MWh, and while the renewable factor has risen slightly to 0.916 the capacity shortage has at the same time stayed close to 0%.

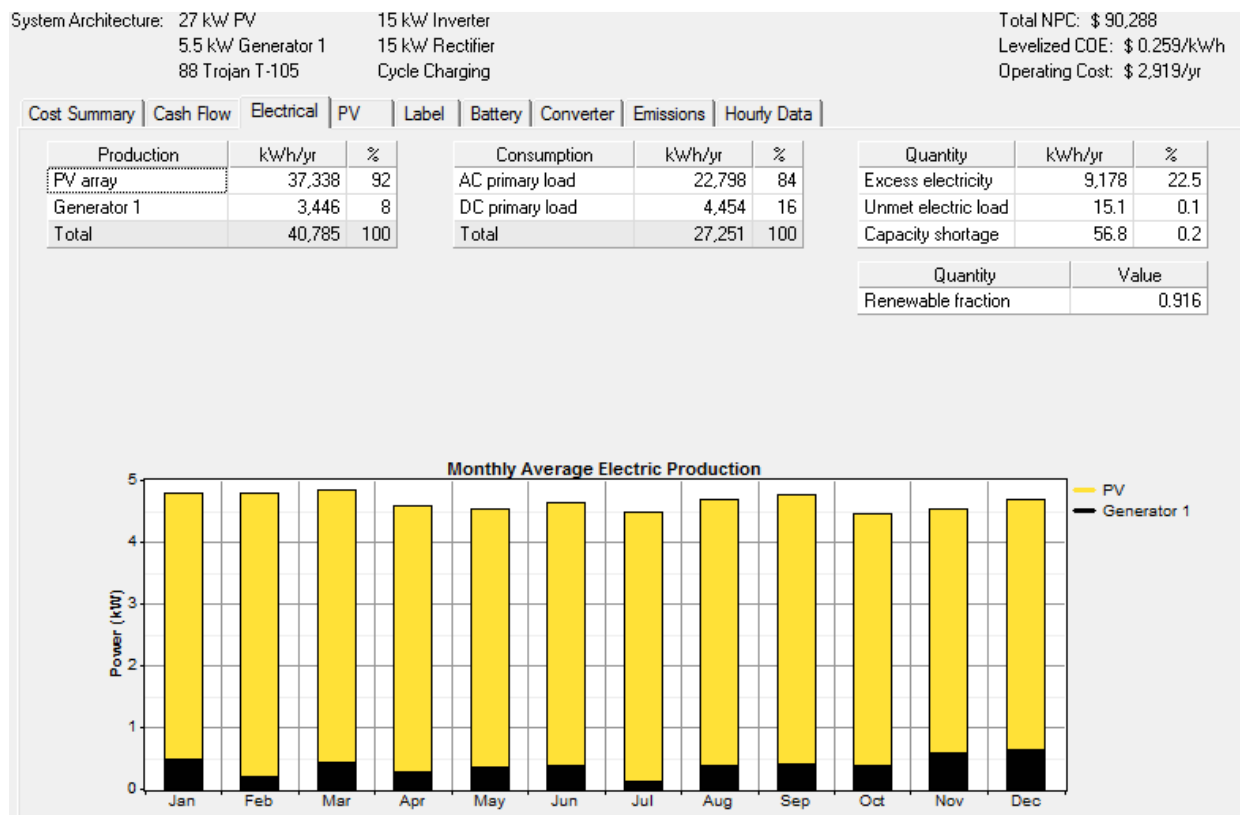


Figure 4.5: Electrical production of the hybrid system.

4.1.5. Conclusion

Compared to the base case hybrid system, the optimized AC/DC PV/hybrid system has smaller battery bank and the PV array size is increased from 26 kW to 27 kW. This results in ~5.2% lower investment costs and ~2% lower net present costs. The levelized cost of electricity is only 1.9% lower.

The simulation of the system was simplistic and didn't take into account the distribution of PV and the DC grids nor the need and costs for additional equipment such as charge controllers and cables. A possibility of lowering the AC load even further would be to have all the lighting and refrigerators in the DC grid. Low voltage DC equipment can be expensive and while having AC and DC plugs in the same area it needs to be taken care that it isn't possible to accidentally damage devices by connecting them to wrong plugs.

For the future it could make sense to have night loads in DC as the losses from the inverters can be avoided. The future system extensions could be done block wise, seeing having distributed power generation would allow shorter cable lengths to provide DC to the night loads and have lower distribution losses. At nighttimes the inverters connecting the blocks to the AC grid could be turned off to avoid their standby consumption.

For the simulation the lamp costs and the extra costs from building DC grid(s) were not included. The price of a security light, whether it was AC or DC lamp can vary a lot. Depending on where they are bought one can find proper security lights both AC and DC around 25 €/lamp so the difference between lamp costs is relatively trivial. However cables and additional components would be required for the DC grids. If the DC grids were to be built it could be economically wise to have more equipment running on DC, such as more lighting, refrigerators and radios. This way lower power consumption, inverter losses and power distribution losses could be achieved. Whether this would be economically wise or not needs further studying.

4.2 Inverter stacking

This subchapter studies the possibility of stacking inverters and its advantages in comparison to centralized inverters. Multiple inverters can be installed in a configuration called stacking.

Stacking refers to the way the inverters are wired within the system and then programmed to coordinate activity. Stacking inverters allows them to work as a single system. One inverter is set as a master and is working the most. Other inverters are set as slave and will provide assistance when the loads are higher than the master inverter is capable of handling. Inverters can be stacked both in single-phase and three-phase. (Davis, 2012.)

4.2.1. Product overview

Steca markets an off grid solution, Steca PI set, which connects up to 4 Steca Solarix PI inverters in parallel to Steca PAX4 parallel switch box totaling in 4.4 kW power. The stacked inverters work as one unit and the advantage of Steca's system is that it is able to shut down the inverters that are not in use. Compared to a larger centralized inverter this type of solution is able to run closer to its nominal power and shutting down the excess inverters is saving energy. Also one good point in the system is that it doesn't rely only on one inverter. In the figure 4.6 is an example of Steca Solarix PI setup with both DC and AC outputs. Up to 4 Steca PAX4 switch boxes can be operated in parallel. (Steca, 2013.)

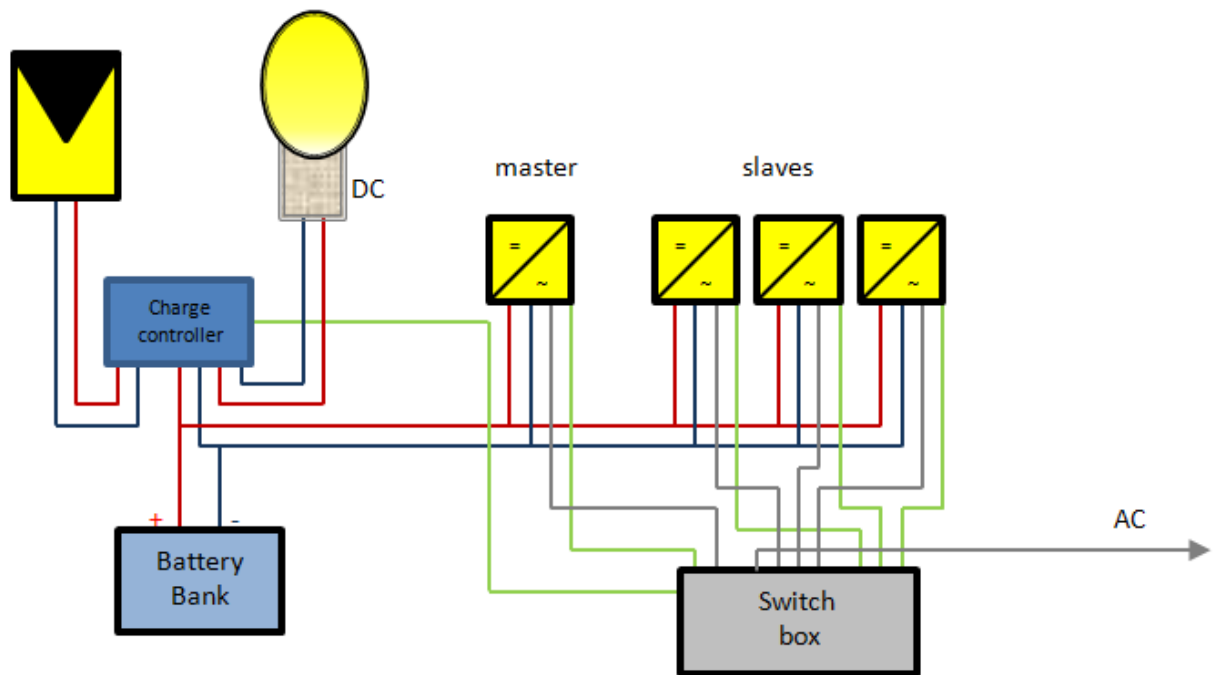


Figure 4.6: Example System layout of a system with Steca Solarix PI inverters in parallel. Notion: Green is the data cable.

OutBack Power Systems have their own option for their inverters called Outback stacking. In parallel stacking the inverter outputs are connected to hot leg 1, in series stacking each inverter output is connected to a separate hot leg. In 3phase stacking the master inverter is set to 3ph Master and the first & second slave are set to 3ph slave. Outback stacking will allow the inverters to act as a single system, charging batteries and providing power to loads. In the system power save levels can be set for the inverters. This will allow the unneeded inverters to sleep when there are low loads. When certain power level is enabled the inverters with matching power save level will wake up. Up to 10 inverters can be stacked totaling in 36 kW power. (OutBack Power Systems, 2013.) Both Steca and OutBack Power Systems offer a quite similar set up. The downside of Steca's system is its small size 4.4 kW although 4 of these should be able to be connected parallel it is still smaller than the largest system OutBack Power Systems has to offer, 36 kW.

Also Victron offers products that can be connected together to achieve higher voltage or power. For instance up to 6 Victron MultiPlus inverter/chargers can be connected in parallel in one phase. Three MultiPlus units can be configured for three phase output and six of these sets can be parallel connected totaling to 72 kW inverter power. MultiPlus inverters have two power saving options. One is automatic economy switch which slightly narrows the sinusoidal wave with low loads and decreases inverter's power consumption approximately 20%. The other option is "search mode" which decreases the power consumption by approximately 70% in low load operation. While "search mode" is selected the MultiPlus will in inverter mode switch off in case of no load and switch on every two seconds for a short period. MultiPlus will continue to operate in inverter mode if the output current exceeds a set level and shut down again if not. (Victron Energy, 2013a.) In figure 4.7 an example of a PV/hybrid system with three parallel connected inverters configured for three phase output is shown. Victron MultiPlus inverter/chargers don't offer the same kind of synchronization as the other two but the inverters can be stacked to achieve as high power as 72 kW and the MultiPlus has also its own means for saving power.

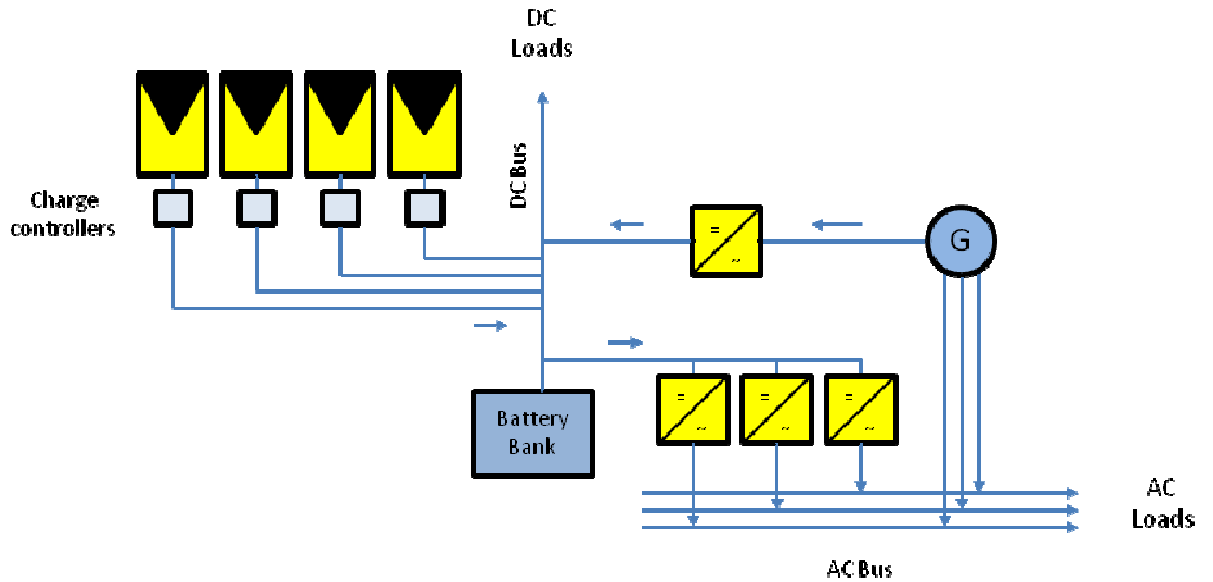


Figure 4.7: System layout of a 3phase PV/Hybrid System with three parallel connected inverters.

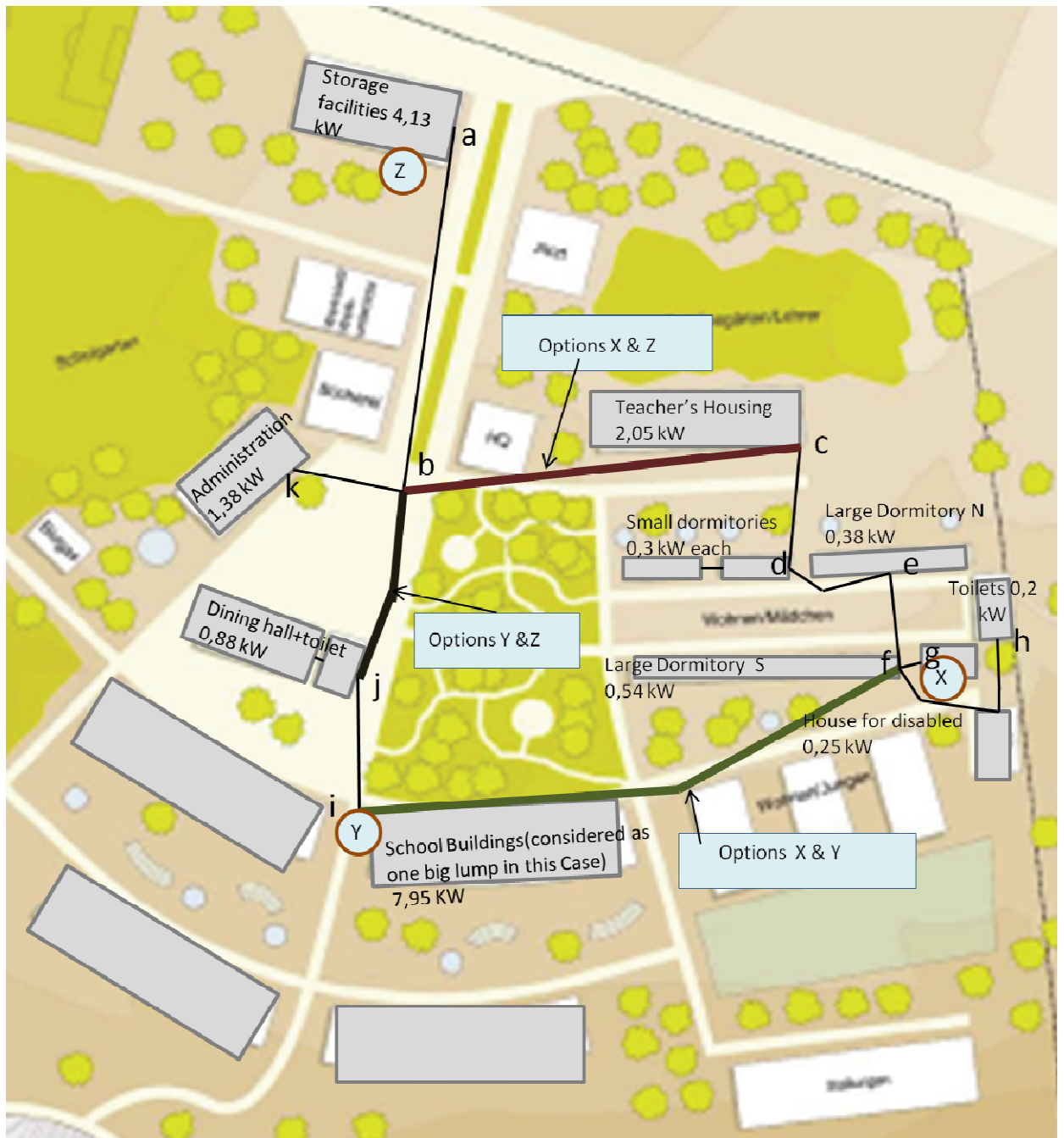
4.2.2. Conclusion

In comparison of having one centralized inverter or connecting multiple inverters together, the latter option is more reliable in the sense that it is not dependent on only one inverter. The overviewed products can also handle varying loads more efficiently than a single large inverter but they are most likely more expensive as well. Using multiple inverters in the system could allow the system to be expanded gradually, inverter by inverter. Inverter stacking could also ensure saving energy and working with high efficiency.

4.3 Centralized power generation – effect of location on cable sizing

In this sub-chapter it is studied how concentrating the power generation in one location and having different 230 VAC circuits effects the wire size. Three different locations, shown on picture 4.2, were chosen for this review with following motivation:

- Location X was chosen as it is close to several buildings and also in the future new dormitories will be build close by. The location has also potentially large night loads and one might wish to use DC for the night load in these buildings.
- Location Y was selected for the large day and night loads the school buildings have, it is also relatively close to the planned new dormitories and to the dining hall and administration building.
- Location Z is reviewed for it has large load and the future buildings for the doctor, library and work shop will be built close to it.



Picture 4.2: Case III master plan. Loads for each building and three possible locations, X(g), Y(i), Z(a) for centralized power generation.

Notion: The positions for connection boxes are marked with small letters. There are three cable sections pointed out that are only used in two of the three options.

4.3.1. Cable sizing

The cable cross section can be calculated with the following formula. For the calculations it is assumed that maximum loss of 3% is accepted (The German Energy Society, 2008, 323). For each cable length between two different connection boxes, pointed out in the picture 4.2, with current estimated load a minimum cable cross section is calculated.

Table 4.2: Electrical parameters for the sizing of cable cross-sections (The German Energy Society, 2008, 323).

Electrical parameters	Symbol	Unit
Cable length	L	m
Power transferred in the cable	P	W
Cable cross section	A	mm ²
Electrical conductivity (copper $k_{CU} = 56$)	k	$\Omega \times \text{mm}^2$
System voltage	V	V

Example calculation for cable f-i option X as the location for centralized power generation.

$$A = L * \frac{P}{3\% * V^2 * k} = 300 \text{ m} * \frac{8830 \text{ W}}{3\% * (230 \text{ V})^2 * 56 \Omega \times \text{mm}^2} = 29.81 \text{ mm}^2$$

Each of the tables, 4.3; 4.4 and 4.5 show the minimum required cable cross section for each cable between given connection boxes. When the required cable cross sections are reviewed, we can see that choosing option X would require fairly long and thick cable between connection boxes f-i (large dormitory S – school buildings). As the loads at the school buildings will get higher in the future the cable would need to be heavily oversized for the current estimated loads. It would also be possible to have more than one cable run between f-i to have the cable cross section smaller.

Table 4.3: Option X: Minimum required cable cross sections for the cable runs.

OPTION X Cable	required cable cross section [mm ²]	Approx. cable length [m]	Connected Load [kW]
X -f	2.07	10	18.35
f-e	5.25	50	9.32
e-d	4.59	50	8.16
d-c	5.95	70	7.56
c-b	12.40	200	5.51
b-k	0.93	60	1.38
b-a	8.83	190	4.13
f-h	0.11	50	0.20
f-i	29.81	300	8.83
i-j	0.595	60	0.88
X-g	0.028	10	0.25

In comparison to option X, Y has the advantage that it doesn't require a long and thick cable to run between the school buildings and the dormitories. As the power generation is located next to the largest load it reduces the cable cross sections in overall. Also as mentioned, the load at school buildings will get higher and this would motivate even further to have power generation close to the school buildings.

Table 4.4: Option Y: Minimum required cable cross sections for the cable runs.

OPTION Y Cable	required cable cross section [mm²]	Approx. cable length [m]	Connected Load [kW]
Y-j	4.32	60	6.39
j-b	6.20	100	5.51
b-k	0.93	60	1.38
b-a	8.83	190	4.13
i-f	13.53	300	4.01
f-h	0.11	50	0.20
f-g	0.03	10	0.25
f-e	1.70	50	3.03
e-d	1.49	50	2.65
d-c	1.61	70	2.05
Y-i	0.90	10	7.95

In option Z long distances to the loads cause the cable Z-b to be long and thick. Having connection box “b” closer to power generation point **Z** would require the rest of the cables to get longer and thicker. This option doesn’t seem to be a good choice for centralizing all the power generation.

Table 4.5: Option Z: Minimum required cable cross sections for the cable runs.

OPTION Z Cable	required cable cross section [mm²]	Approx. cable length [m]	Connected Load [kW]
Z-b	30.40	190	14.22
b-k	0.93	60	1.38
b-j	9.94	100	8.83
j-i	5.37	60	7.95
b-c	9.02	200	4.01
c-d	1.55	70	1.96
d-e	0.77	50	1.36
e-f	0.55	50	0.98
f-g	0.03	10	0.25
f-h	0.11	50	0.20
Z-a	0.47	10	4.13

4.3.2. Conclusion

In centralized power generation it would make sense to have the largest loads nearby to avoid the need for long and thick cables. The future loads should also be taken into consideration and for this, out of these options the centralized power generation should be either close to the school buildings at point Y or to the dormitories at point X. School

buildings will have higher loads, but depending on the future system design also the diesel generator's location should be thought thoroughly as in daytime it could cause noise pollution in the school area. Then again, a loud diesel generator should neither be located next to the dormitories.

5 Discussion and conclusions

From the optimized base case systems the PV/hybrid system was the most reasonable in term of reliability and costs. This result was in line with the previous studies where in fairly similar weather conditions and amounts of solar radiation PV/hybrid systems have been found as a good choice.

In comparison to the study on decentralized power generation with hybrid systems in Bangladesh by Mondal & Denich there were some major system sizing differences between the system optimizations. The optimized base case hybrid system for the educational center has 26 kW PV; 5.5 kW diesel generator and 151 kWh battery capacity to supply 75 kWh/d with 16 kW peak load while their optimal system, for supplying 50 kWh/d load with a 11 kW peak load, consisted of a 6 kW PV; 10 kW diesel generator and a 16 kWh battery bank. One of the major factor explaining their system's high reliance on the diesel generator is the low fuel price 0.65 \$/l (Mondal & Denich 2010) which costs significantly less than the 0.94 €/l used in the optimization of the base case system for the educational center. The low fuel cost makes running the diesel generator in their system economically more sense than investing in expensive batteries and PV modules.

The results achieved in this study are slightly different to the study made by Givler & Lilienthal (2005), where they found out that “the PV/battery systems were most cost-effective up to loads ranging from 3 kWh/d to 13 kWh/d depending on the reliability, solar resource and diesel fuel price. The optimization done in this paper's chapter 3 suggests that the PV/battery system is the optimal choice even with the 75 kWh/d consumption. The simulation results in this paper didn't take the reliability into consideration, which was one of the reasons why this paper suggests a PV/hybrid (PV/battery/diesel generator) system over a PV/battery system. The system reliability is something also Rohit Sen (2011) in his study pointed out and underlined the importance of having more than one mean of power generation. In the study by Givler & Lilienthal (2005), the fuel (diesel) costs were very low, only 0.5 \$/l also their costs for batteries were lower than in this study. Their investment costs for PV generation was ~7800\$/kW which seems very expensive. Together these factors would motivate them to have a PV/diesel hybrid system even at lower daily loads.

Givler & Lilienthal (2005) also studied the effect of allowing a small capacity shortage factor and found out it made the PV/battery system more competitive. This was also the case in the simulations in this paper. If no shortage capacity factor would have been accepted the optimized PV/battery system for the chapter 3 base case system would have been 17000 € more expensive than the PV/hybrid system. The PV/battery system without any shortage capacity factor would have required 39 kW PV, 20kW converter and 184 T-105 batteries. With the allowed 5% capacity shortage factor, the PV/battery system is 9000 € cheaper than the PV/hybrid system.

In both chapter 3 and subchapter 4.1 the optimization suggested the system to have 15 kW converter capacity. However the PV array sizes in these systems are larger than that. It could be that HOMER assumes this 15 kW AC output combined with diesel generators output is enough to supply the peak loads. The “excess PV generation” that is not converted into AC is most likely used for charging batteries and in subchapter 4.1 for the DC grid. Converting the whole supply to AC does make little sense because high currents

are drawn only for short periods, thus for most of the time inverter would be working in lower partial load and lower efficiency (The German Energy Society, 2008). For instance the Victron MultiPlus inverters used in the base case system in chapter 3 (3 x 5 kW) would each be capable of 10 kW peak inverter output for a short period. For these reasons larger converter wasn't chosen. The AC loads in future will surely get larger and the conversion capacity might eventually have to be larger as well. Inverter stacking could offer a solution for increasing the conversion capacity gradually. One just has to remember to make sure the units are compatible and support stacking. The future loads will also need to be considered in the power generation system design. As was seen in the subchapter 4.3 a proper location selection for a centralized system can help to avoid long thick cable runs that are expensive. The school buildings have high daytime power consumption compared to the rest of the educational center and for that the power generation ought to be relatively close to it. The noise level of the diesel generators should also be considered when deciding locations.

In subchapter 4.1 different optimization possibilities were studied. It was shown that having DC night loads could lower the costs and the systems battery size would be smaller. In future it could be worthwhile to invest in DC equipment, such as lighting and refrigerator to have lower consumption and to avoid some DC/AC conversion losses. It would also make sense to have distributed power generation system where extension would be made block wise. This would enable DC supply for buildings close to the power generation and having the power generation close to the loads will keep the cable sizes smaller. The disadvantage of having DC grids is the costs for all the additional components and cabling. The higher the load connected into DC grids the more having DC grids will make sense. The system extension in this type of system could be done by connecting power generation units to the AC grid via bi-directional inverters.

For centralized power generation, the inverter stacking where several inverters can be connected together to work as a one unit with high efficiency and being able to shut down the excess inverters could be proper option to consider. Since there will be significant varying in the loads and in comparison of running one large centralized inverter the efficiency and reliability of stacked inverters would be higher. This system could be extended by adding more inverters either parallel or series depending on what is required in order to add more PV generation.

An example of a possible system that would combine the advantages of inverter stacking and DC night loads is shown in picture 4.1 where the inverter at point A could be considered as multiple stacked inverters.

One major factor affecting the system size is the night consumption. By lowering the night consumption it would be possible to reduce the system size and costs. It should be further investigated if the night loads could be reduced by choosing security lights with lower power consumption. The estimated power consumption for each security light is 20 W. It could make sense to invest slightly more and have DC refrigerators instead of AC. Even though they can be more expensive, night consumption could be reduced by avoiding power conversion losses.

Carefully scheduling tasks with high power consumption will help keeping the system size smaller. Majority of the power consumption should be spread across daytime when PV generation is available and unnecessary night loads should be avoided as they rely on battery capacity and the diesel generator.

All the simulation results achieved can and will change if the inputs are changed. For this reason none of the results should be taken as carved in stone. Rather the results are only indicative. Assumptions made and input parameters have been motivated earlier in other chapters.

6 References

Al-Badi, A.H, AL-Toobi,M , AL-Harthy, S. , Al-Hosni, Z. & AL-Harthy, A. (2012) Hybrid systems for decentralized power generation in Oman. *International Journal of Sustainable Energy*, 31:6, 411-421.

Bahati, C. (2010) Mavuno Project: Construction of a Secondary School in Chonyonyo. Progress Report, July 2010.

Bahati, C. (2012) Mavuno Project: Construction of a Secondary School in Chonyonyo. Progress Report.

Birol, F. (2007) Energy Economics: A Place for Energy Poverty in the Agenda?. *The Energy Journal; International Journal of Energy Research*, 28, 3, 4-5.

Central Intelligence Agency (2013) The World Factbook:Tanzania. [Online] Available from: <https://www.cia.gov/library/publications/the-world-factbook/geos/tz.html> [Accessed 29/05/13].

Central Maine Diesel (2013) Automatic diesel generator. [Online] Available from: http://www.generatorsales.com/order/Automatic-6000-Watt-Diesel-Generator2.asp?page=AUTO_DIESEL [Accessed 11/05/2013].

Davis,S. (2012) Inverter/Charger Supports Grid-Interactive and Off-Grid Applications. [Online] Available from: <http://powerelectronics.com/inverters/invertercharger-supports-grid-interactive-and-grid-applications> [Accessed 27/05/13].

Givler, T. & Lilienthal, P. (2005) Using HOMER® Software, NREL's Micropower Optimization Model, to Explore the Role of Gen-sets in Small Solar Power Systems; Case Study: Sri Lanka. *Technical Report* NREL/TP-710-36774 May 2005 Colorado:National Renewable Energy Laboratory

Haanyika, C. M. (2005) Rural electrification policy and institutional linkages. *Energy Policy* 34 (2006) 2977-2993.

Holliman, J. (2010) Designing a Photovoltaic System Hybrid for a Health Clinic in Camilo Ortega, Nicaragua. San Luis Obispo: Electrical Engineering Department California Polytechnic State University.

Ingenjörer utan gränser (2013) Elsystemlösningar för utbildningscenter i Tanzania [Online] Available from: <http://ingenjorerutangranser.se/projekt/elsysteml%C3%B6sningar-f%C3%B6r-utbildningscenter-i-tanzania> [Accessed 29/05/13].

International Energy Agency (2011), World Energy Outlook 2011. [Online] Available from: <http://www.worldenergyoutlook.org/publications/> [Accessed 01/05/13.]

International Energy Agency (2010), World Energy Outlook 2010. [Online] Available from: <http://www.worldenergyoutlook.org/publications/> [Accessed 01/05/13.]

IOL (2012) Tanzania hikes petrol, diesel prices. [Online] Available from: <http://www.iol.co.za/business/business-news/tanzania-hikes-petrol-diesel-prices-1.1435433> [Accessed 24/03/13].

- Klimstra, J. (2012) Africa yearns for electricity. [Online] Available from: <http://www.wartsila.com/fi/africa-yearns-for-electricity> [Accessed 29/05/13].
- Markvart, T (2000) Solar Electricity 2nd edition. Chichester: John Wiley & Sons.
- Mavuno Project (2013) [Online] Available from: <http://www.mavunoproject.org/> [Accessed 24/03/13]
- Mavuno Project (2013) programs [Online] Available from: <http://www.mavunoproject.org/projects.html> [Accessed 24/03/13].
- Mondal, A.H & Denich, M. (2010) Hybrid systems for decentralized power generation in Bangladesh. Energy for Sustainable Development 14 (2010) 48–55
- OutBack Power Systems (2013) Stacking explained. [Online] Available from: http://www.outbackpower.com/pdf/tech_notes/Stacking_Explained.pdf [Accessed 14/05/13].
- Phaesun (2013) Hybrid Systems. [Online] Available from: <http://www.phaesun.com/systems/references/rural-electrification/hybrid-systems.html> [Accessed 30/05/13].
- Sen, R. (2011) Off-Grid Electricity Generation with Renewable Energy Technologies in India; An application of HOMER. MSc. Renewable Energy & Environmental Modelling University of Dundee.
- Steca (2013) Pv off grid - product catalogue. [Online] Available from: http://www.steca.com/index.php?PV_off_grid [Accessed 14/05/13].
- Sustainable.co.za (2013) Victron MultiPlus 5 kW Inverter/Charger. [Online] Available from: <http://www.sustainable.co.za/victron-multiplus-5000w-pure-sinewave-inverter-charger-solar-inverter.html> [Accessed 01/05/2013].
- Sustainable.co.za (2013a) Victron Trina Solar 230 W Solar Panel. [Online] Available from: <http://www.sustainable.co.za/trina-solar-tsm-pc05-230w-pv-module-solar-panel.html> [Accessed 01/05/2013].
- Sustainable.co.za (2013b) Victron Trojan T-105 Battery. [Online] Available from: <http://www.sustainable.co.za/trojan-re-series-t105-225ah-6v-solar-storage-batteries.html> [Accessed 01/05/2013].
- The German Energy Society (2008) Planning & Installing Photovoltaic Systems: A guide for installers, architects and engineers 2nd edition. London: Earthscan.
- The Government of the United Republic of Tanzania. (2004) Gender. [Online] Available from: <http://www.tanzania.go.tz/gender.html> [Accessed 24/03/13].
- Trojan Battery Company (2013) T-105 data sheet. [Online] Available from: http://www.trojanbatteryre.com/PDF/datasheets/T105_TrojanRE_Data_Sheets.pdf [Accessed 11/05/13].
- Trina Solar (2012) TSM .PC 05 data sheet. [Online] Available from: http://www.trinasolar.com/HtmlData/downloads/us/products/mono/PC05_Datasheet_Dec12_EN.pdf [Accessed 11/05/13].

Victron Energy (2013) Multiplus inverter/charger 800Va – 5kVa data sheet. [Online] Available from: <http://www.victronenergy.com/inverters-chargers/multiplus-12v-24v-48v-800va-3kva/> [Accessed 11/05/2013].

Victron Energy (2013a) MultiPlus Manual. [Online] Available from: <http://www.victronenergy.com/inverters-chargers/multiplus-12v-24v-48v-800va-3kva/> [Accessed 14/05/03].

World Bank Group (2010). Addressing the Electricity Access Gap. [Online] Available from: http://siteresources.worldbank.org/EXTESC/Resources/Addressing_the_Electricity_Access_Gap.pdf [Accessed 30/05/13].

World Health Organization (2010) WHO guidelines for indoor air quality: Selected Pollutants. Bonn: The WHO European Centre for Environment and Health.

Appendix A: Load profile for the educational center

		Power / piece	Number	Total Power [W]	0-1	1-2	2-3	3-4	4-5	5-6	6-7	7-8	8-9	9-10	10-11	11-12	12-13	13-14	14-15	15-16	16-17	17-18	18-19	19-20	20-21	21-22	22-23	23-24	Sum total [Wh]
Large Dormitories	Light	20	36	720																				720	720	720			2160
	Security light	20	8	160	160	160	160	160	160	160	160													160	160	160	160	160	1920
Small Dormitories	Light	10	32	320																				320	320	320			960
	Light	6	16	96																				96	96	96			288
	Light by tanks	6	4	24																				24	24	24			72
	Security light	20	8	160	160	160	160	160	160	160	160													160	160	160	160	160	1920
House for Disabled	Light	10	5	50																				50	50	50			150
	Light	6	4	24																				24	24	24			72
	Security light	20	2	40	40	40	40	40	40	40	40													40	40	40	40	40	480
	Fridge	100	1	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	2400
	Laptop	30	1	30														30	30										60
	Charging phones	2	1	2															4	4	4								12
Bathrooms and Toilet	Light	6	40	240																				240	240	240			720
	Security light	20	6	120	120	120	120	120	120	120	120													120	120	120	120	120	1440
Class rooms (6)	Light	40	24	960								960	960	960	640	320	320	640	640	640	320								6400
	Beamer	250	1	250											250	250													500
	Laptop	30	1	30								30	30	30	30	30													150
	Security light	20	12	240	240	240	240	240	240	240	240													240	240	240	240	240	2880
Computer room	Light	40	4	160													160	160											320
	Computers	300	20	6000													4000	6000											10000
	Beamer	250	1	250													250	250											500
	Printer	20	1	20													4	4											8
	Security light	20	2	40	40	40	40	40	40	40	40													40	40	40	40	40	480
Administration Block	Light	20	8	160								160	160	160	160	160	160	160	160	160	160								1600
	Computers	100	10	1000								500	1000	1000	800	400	800	1000	1000	600	500								7600
	Printer	20	1	20								4	4	4	4	4	4	4	4	4	4								40
	Scanner	20	1	20								4	4	4	4	4	4	4	4	4	4								40
	Copy machine	20	1	20								4	4	4	4	4	4	4	4	4	4								40
	Fridge	100	1	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	2400
	Charge Phones	2	10	20								4	4	4	4	4	4	4	4	4	4								40
	Security light	20	2	40	40	40	40	40	40	40	40													40	40	40	40	40	480
Dining Hall	Light	40	10	400							400	400	400	400	400	400	400	400	400	400	400	400	400						5200
	Music system	60	1	60							60	60	60	60	60	60	60	60	60	60	60	60	60						780
	TV	100	1	100																	100	100	100	100					400
	Receiver	5	1	5																	5	5	5	5					20
	DVD player	12	1	12																	12	12	12	12					48
	Security light	20	4	80	80	80	80	80	80	80	80													80	80	80	80	80	960
Storage Facilities	Light, big hall	20	10	200								200	200	200	200	200	200	200	200										1400
	Milling machine	3500	1	3500											3500	3500													7000
	Sealing machine	350	1	350													350	350	350	350									1400
	Security light	20	4	80	80	80	80	80	80	80	80													80	80	80	80	80	960
Kitchen	Light	20	4	80							80	80	80	80	80	80	80	80											720
	Security light	20	1	20	20	20	20	20	20	20	20													20	20	20	20	20	240
Teacher's houses	Various items	500	10	5000																									0
	radio	10	3	30							10	30	30							20	20	30	30	30	20	20			240
	Receiver	5	1	5							5	5	5						5	5	5	5	5	5	5	5			50
	tv	100	1	100							100	100	100						100	100	100	100	100	100	100	100			1000
	laptop	30	6	180							90	150	150						180	180	180	150	120	90					1290
	phone	4	3	12							12	12	12						12	12	12	12	12						96
	cooking devices	1500	1	1500							300	300							1500	900	900	500							4400
	fridge	100	1	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	2400
	light	20	4	80																				80	80	80			240
	security light	20	2	40	40	40	40	40	40	40	40													40	40	40	40	40	480
Load Sum					1320	1320	1320	1320	1320	1400	2377	3073	3503	3206	6436	6066	7130	9620	3156	2644	3634	2108	2074	3641	3089	2999	1320	1320	75456